Washington University in St. Louis

presenting

psyduck

AIAA DBF 2025-2026 Design Report

Design Build Fly

washu

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Nomenclature

Symbol	Definition
α	Angle of Attack
ν	Kinematic Viscosity
ρ	Air Density
λ	Taper Ratio
A	Main Wing Area
b	Main Wing Span
b_{HT}	Horizontal Tail Span
b_{VT}	Vertical Tail Span
C_C	Cargo Operating Cost Per Lap
C_D	Coefficient of Drag
C_e	Base Operating Cost Per Lap
C_L	Coefficient of Lift
$C_{l_{\delta_\alpha}}$	Aileron Authority
$c_{l_{\delta_\alpha}}$	Change in Lift Coefficient with Aileron Deflection
C_m	Coefficient of Moment
C_P	Passenger Operating Cost Per Lap
c_R	Root Chord
c_{ref}	Geometric Chord
D	Drag
D_f	Skin Friction Drag
D_p	Pressure Drag
D_t	Total Drag
F_D	Drag Force
F_L	Lift Force
$Ic1$	Fixed Income Per Cargo
$Ic2$	Income Per Cargo Per Lap
$Ip1$	Fixed Income Per Passenger
$Ip2$	Income Per Passenger Per Lap
I_T	Optimal Arm Length
L	Lift
N_{cargo}	Number of Cargo
N_{laps}	Number of Laps
$N_{passenegers}$	Number of Passengers
S_{HT}	Horizontal Tail Area
S_{ref}	Main Wing Area
S_{VT}	Vertical Tail Area
T/W	Thrust-to-Weight Ratio
u	Cruise Velocity
V_∞	Freestream Velocity
V_{HT}	Horizontal Tail Volume
V_{VT}	Vertical Tail Volume
X_{CG}	CG X-location
X_{NP}	Neutral Point X-location

Abbreviation	Definition
AIAA	American Institute of Aeronautics and Astronautics
AR	Aspect Ratio
AUW	All Up Weight
BEC	Battery Eliminator Circuit
CAD	Computer-Aided Design
CF	Carbon Fiber
CFD	Computational Fluid Dynamics
CG	Center-of-Gravity
COTS	Commercial off-the-shelf
DBF	Design/Build/Fly
EF	Efficiency Factor
ESC	Electronic Speed Controller
FEA	Finite Element Analysis
GM	Ground Mission
HT	Horizontal Tail
LiPo	Lithium Polymer
M1	Mission 1
M2	Mission 2
M3	Mission 3
MAC	Mean Aerodynamic Chord
MTOW	Maximum Takeoff Weight
PETG	Polyethylene Terephthalate Glycol
PLA	Polylactic Acid
RAC	Rated Airplane Cost
Re	Reynolds Number
SM	Static Margin
SPN	Servo Power Node
VLM	Vortex Lattice Method
VT	Vertical Tail
WashU	Washington University in St. Louis
Wh	Watt-hours
WUDBF	Washington University in St. Louis Design Build Fly

1 Executive Summary

1.1 Introduction and Mission Requirements

The Washington University in St. Louis Design/Build/Fly (WUDBF) 2025–26 Design Report presents the development of *Psyduck*, the team's entry in the American Institute of Aeronautics and Astronautics (AIAA) 2026 Design-Build-Fly (DBF) competition. The contest challenges teams to design a banner-towing bush plane capable of completing three sequential flight missions and one ground mission. Mission 1 (M1) demonstrates baseline airworthiness and controllability. Mission 2 (M2) tasks the aircraft with transporting a selectable combination of passengers and cargo to maximize laps flown within a five-minute flight window. Mission 3 (M3) requires the aircraft to deploy, tow, and jettison a banner while maximizing banner length and laps within a five-minute flight window. The Ground Mission (GM) further evaluates the aircraft's ability to rapidly transition between charter and banner-towing configurations and deploy and jettison on the ground.

1.2 Development Process Summary

The development of *Psyduck* followed a structured, mission-driven design process consisting of conceptual, preliminary, and detailed design phases. Conceptual design, informed by trade studies, optimization, sensitivity analysis, and performance insights from past WUDBF aircrafts, established a conventional-tail, low-wing configuration with a single tractor motor and bow-tricycle landing gear. This layout optimizes ground-handling stability, payload accessibility, and banner-deployment requirements. Preliminary design quantified aerodynamic and propulsion performance through lift, drag, stability, and power analyses. Analyses informed selections for wing loading, tail sizing, center-of-gravity (CG) range, banner design, payload, and mechanism integration. Detailed design refined the internal structure, payload-restraint systems, and banner-deployment mechanism while integrating propulsion and electronics systems to ensure reliable operation across all missions. At the conclusion of this process, the finalized design features a wingspan (WS) of 4.59 ft, overall length of 4.70 ft, maximum height of 1.67 ft, a passenger capacity of 16 ducks, a cargo capacity of 4 pucks, a banner length of 60 in, and a propulsion battery capacity of 82.22 Watt-hours (Wh).

1.3 Aircraft Predicted Performance

As shown in Figure 1.1, the finalized aircraft geometry supports the performance objectives of all competition missions. The aerodynamic design enables strong low-speed handling suitable for charter operations and stable flight during high-drag banner towing. The propulsion system provides sufficient thrust and endurance to complete the five-minute mission windows with margin. Internally, the payload architecture allows rapid loading during staging and ensures secure restraint of passengers, cargo, and the stowed banner throughout flight operations. Table 1.1 summarizes the expected performance metrics and mission-level capabilities, demonstrating that the aircraft is well positioned to meet the competitive demands of the 2026 DBF competition.

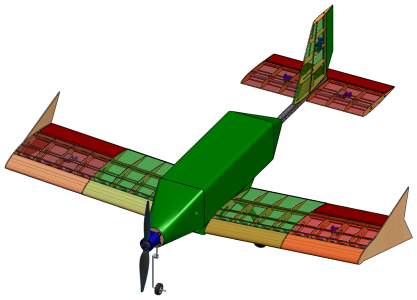


Figure 1.1: Final Aircraft Design

Table 1.1: Psyduck Predicted Performance

Parameter	M1	M2	M3	GM	Total
Gross Weight (lbs)	7.68	10.88	8.08	-	-
Payload Details	-	16 passengers 4 cargo	60 in Banner	-	-
Number of Laps	3	8	6	-	-
Mission Time (min)	1.81	4.82	4.65	1.62	-
Cruise Speed (ft/s)	115	114	92	-	-
Predicted Mission Score	**1.00**	**1.62**	**2.46**	**0.39**	**5.47**

2 Management Summary

2.1 Organization

WUDBF is organized into two primary branches: technical and administrative. The technical branch is divided into six sub-teams, while the administrative branch consists of five distinct executive roles, as shown in Figure 2.1.

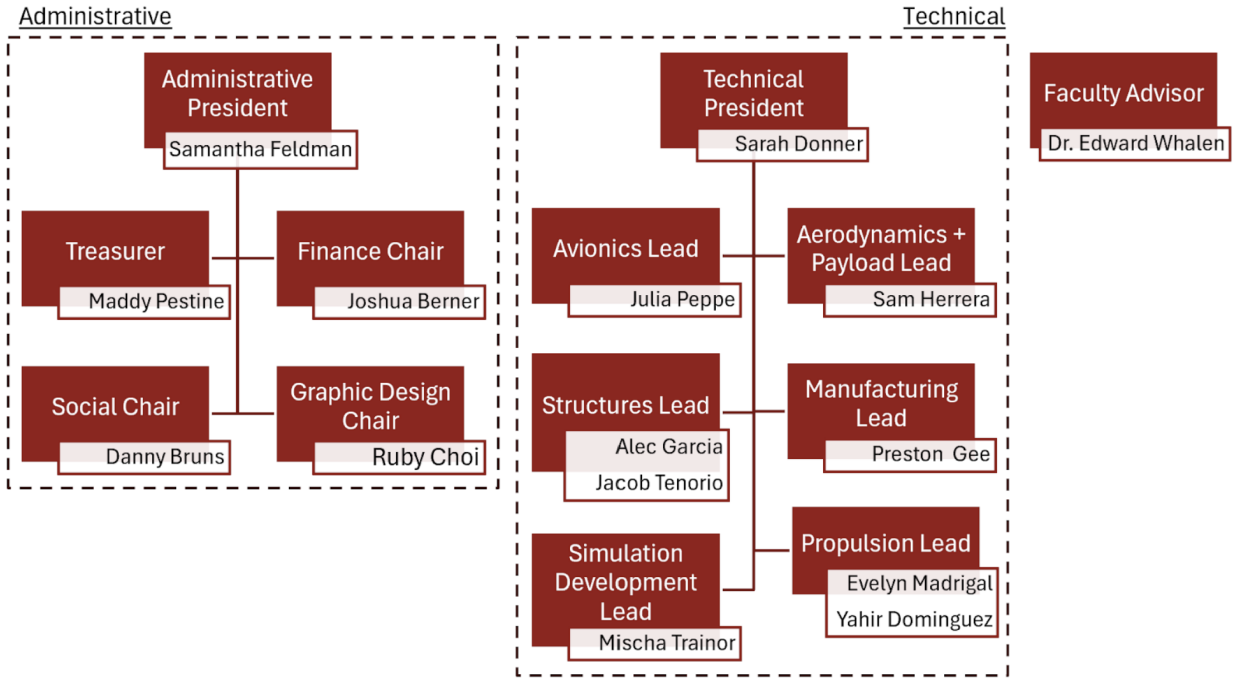


Figure 2.1: Team Hierarchy

The administrative branch operates under the administrative president, managing recruitment, finances, social events, and communication with the Washington University in St. Louis (WashU) community and external partners. Each administrative executive works individually outside of meetings and comes together once a week at WUDBF's general body meetings to relay updates to the administrative president and branch.

The technical president leads the technical branch, which is responsible for designing and engineering the aircraft, training new members, and completing all competition deliverables. Each of the six sub-teams is managed by a team lead, who serves as the subject-matter expert responsible for the final engineering decisions within their subsystem. Team leads plan and run sub-team meetings at the weekly general body meetings. They allocate responsibilities to members based on technical skill, interest, and experience, providing training and guidance to ensure tasks progress. Table 2.1 outlines the roles and responsibilities of each technical sub-team.

The technical branch confers regularly with the faculty advisor to address complex engineering questions, including aerodynamic analysis, design trade studies, and system integration. These meetings provide an opportunity to discuss design concepts and collaboratively brainstorm technical solutions.

Significant technical challenges and cross-functional design decisions are escalated to the leadership board, consisting of presidents, administrative executives, and technical sub-team leads. The leadership board assembles twice weekly to determine collective solutions and define next steps for the design.

Table 2.1: Sub-Team Roles and Skills

	Aerodynamics + Payload	Structures	Manufacturing	Avionics	Propulsion	Simulation Development
Roles	Determine size and optimize wing and tail geometry for stability; design and model mission payloads (banner and release mechanism).	Design, model, and structurally analyze the wing, tail, and fuselage.	Select and analyze materials and assemble aircraft components according to design drawings.	Integrate and install electrical systems (servos, receiver, DAQ); conduct ground and flight data collection and analysis.	Design, select, and test the propulsion system (motor, propeller, battery).	Develop and utilize simulation tools for design optimization and sensitivity analysis.
Skills	Knowledge of aircraft performance, stability, and control theory. CFD (Ansys), XFLR5, CAD (Solidworks).	Knowledge in Solid Mechanics, Statics, Dynamics, CAD (Solidworks), structural FEA, and mass properties analysis.	Knowledge in Materials, laser cutting, fiberglass layups, hot-wire foam cutting, 3D printing, and balsa aircraft construction.	Knowledge in electrical circuitry/PCB design, soldering, Real Flight 8, data collection, and autonomous flight programming.	Knowledge in propeller propulsion theory, Ecalc, and Remote Control (RC) Benchmark.	Knowledge in MATLAB, Simulink, and Python.

2.2 Schedule

WUDBF uses a Gantt chart to ensure the team stays on track to meet major deadlines. The Gantt chart for the 2025-2026 competition is shown in Fig. 2.2 below.

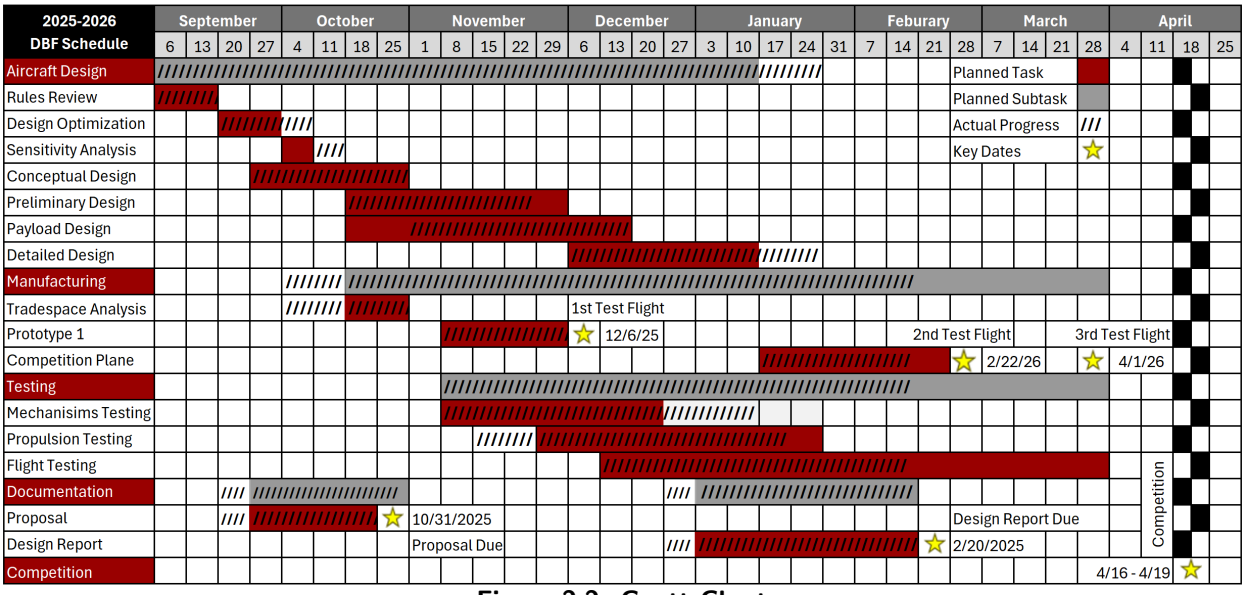


Figure 2.2: Gantt Chart

3 Conceptual Design

3.1 Problem Statement

For the 2026 DBF competition, AIAA challenges teams to design a banner-towing bush plane capable of transporting passengers and cargo while maintaining a high-scoring, stable, and manufacturable configuration. The aircraft must store passengers and cargo, and deploy, tow, and jettison a banner in flight, while demonstrating controlled takeoff, maneuvering, and landing performance throughout all missions. Guided by the competition rules, WUDBF initiated the conceptual design process by defining key mission objectives, translating them into engineering requirements, and conducting scoring sensitivity analyses and configuration trade studies to select an optimal aircraft architecture.

3.2 Requirements and Score Summary

The team's final competition ranking is determined by the total competition score. The total competition score defined by Equation 3.1 is a function of the total report score, total mission score, and participation score.

$$Competition\ Score\ =\ Total\ Report\ Score\ \times\ Total\ Mission\ Score\ +Participation\ Score \tag{3.1}$$

The team's total report score defined by Equation 3.2 is the sum of the proposal score and the design report score, weighted by 15% and 85%, respectively.

$$Total\ Report\ Score\ =0.15\ \times Proposal\ Score\ +\ 0.85\ \times Design\ Report\ Score \tag{3.2}$$

The team's total mission score defined by Equation 3.3 is the sum of all three flight missions and the GM.

$$Total\ Mission\ Score\ =M1\ +M2\ +M3\ +GM \tag{3.3}$$

The team's participation score is given by Table 3.1, where the score is determined by varied participation.

Table 3.1: Aircraft Structural Requirements

Score	Participation
1	Attending the Fly-off
2	Completing Tech Inspection
3	Attempting a Flight Mission

All flight missions follow the flight course given in Figure 3.1. The course consists of two 1000 ft straight segments, two 180° banked turns, and one 360° banked loop. A lap is completed when the aircraft crosses the finish/start line. A successful landing is required to complete each flight mission.

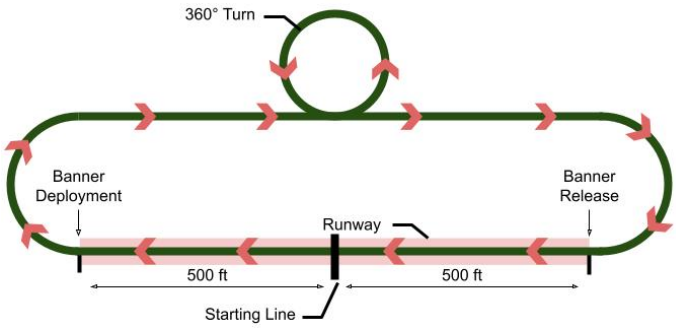


Figure 3.1: Flight Course

3.2.1 Mission 1

M1 is a test flight in which three laps are flown within a five-minute window without any payload. Upon completion, one point is awarded to the team as shown in Equation 3.4.

$$M1 = 1.0 \tag{3.4}$$

3.2.2 Mission 2

M2 simulates a commercial charter flight in which teams must maximize passenger and cargo transport and lap count while minimizing propulsion battery capacity within a five-minute window. The mission scoring is based on net income, which accounts for both revenue generated from transporting passengers and cargo, and the operational costs associated with completing laps. The net income is calculated according to Equation 3.5.

$$Net\ Income = Income - Cost \tag{3.5}$$

Income is calculated by the number of passengers and cargo carried, the number of laps completed, and fixed per-passenger and per-cargo rates, as defined in Equation 3.6. Table 3.2 gives the associated income variables.

$$Income = N_{\text{passengers}} \cdot (I_{p1} + I_{p2} \cdot N_{\text{laps}}) + N_{\text{cargo}} \cdot (I_{c1} + I_{c2} \cdot N_{\text{laps}}) \tag{3.6}$$

Table 3.2: Income Variable Definitions and Values

Variable	Definition	Value
Ip1	Fixed income per passenger	6
Ip2	Income per Passenger per lap	2
Ic1	Fixed income per cargo	10
Ic2	Income per cargo per lap	8

Cost is calculated based on the number of passengers and cargo carried and associated cost variables, scaled by the number of laps completed and the aircraft's propulsion battery capacity through the Efficiency Factor (EF), as defined in Equations 3.7 and 3.8. The corresponding cost variables are summarized in Table 3.3.

$$Cost = N_{\text{laps}} \cdot \left(C_e + N_{\text{passengers}} \cdot C_p + N_{\text{cargo}} \cdot C_c\right) \cdot EF \tag{3.7}$$

$$EF = \frac{Battery\ Capacity}{100} \tag{3.8}$$

Table 3.3: Cost Variable Definitions and Values

Variable	Definition	Value
Ce	Base operating cost per lap	10
Cp	Passenger operating cost per lap	0.5
Cc	Cargo operating cost per lap	2

The final M2 score is calculated by awarding one point for completion plus a fractionalized contribution of net income normalized by the maximum net income achieved by any team, defined in Equation 3.9.

$$M2 = 1 + \frac{Net\ Income}{Net\ Income_{\text{max}}} \tag{3.9}$$

The maximum possible score for M2 is **2.0** points.

3.2.3 Mission 3

M3 is a banner flight in which the banner payload is remotely deployed after the first upwind turn and jettisoned after crossing the finish line. In M3, teams must maximize banner length and laps flown in a five-minute window while minimizing WS. The M3 score is based on the number of laps completed, the banner length, and the Rated Airplane Cost (RAC). The RAC is defined in Equation 3.10 and is used to calculate the M3 score in Equation 3.11.

$$RAC = 0.05 \cdot \left(\frac{WS}{12}\right) + 0.75 \tag{3.10}$$

Where WS is measured in inches and converted to feet within the equation. WS is rounded to the nearest half

inch prior to evaluation, and the minimum allowable RAC is 0.9.

$$M3 = 2 + \frac{\frac{N_{laps} \cdot \text{Banner Length}}{RAC}}{\left(\frac{N_{laps} \cdot \text{Banner Length}}{RAC}\right)_{\text{max}}} \tag{3.11}$$

The final M3 score is computed by awarding two points for successful mission completion, plus a fractional contribution equal to the team's flight performance normalized by the maximum M3 flight score achieved by any team. The maximum possible score for M3 is **3.0** points.

3.2.4 Ground Mission

The GM is a timed demonstration of the team's ability to equip the aircraft for the charter and banner flight missions. The mission begins with the aircraft in the 10 ft $\times$ 10 ft mission box in flight configuration, with all payloads uninstalled. One ground crew member loads and unloads the declared passengers and cargo and installs the banner. The pilot then demonstrates control functionality and, with the aircraft in the vertical position, verifies successful banner deployment and jettison. The final GM score, defined in Equation 3.12, is calculated as the team's mission time normalized by the minimum mission time achieved by any team:

$$GM = \frac{time_{\text{min}}}{time} \tag{3.12}$$

The maximum possible score for GM is **1.0** point.

3.2.5 General Aircraft and Operational Requirements

The requirements presented in Tables 3.4 and 3.5 outline the structural and operational criteria from the competition rules [1]. These requirements form the foundation of the aircraft's design, ensuring regulatory compliance while guiding decisions related to configuration, performance, and safety.

Table 3.4: Aircraft Structural Requirements

Category	Requirement	Req. No.
Configuration	Maximum WS = 5 ft	GR.1
	Aircraft can be any configuration other than rotary-wing or lighter-than-air	GR.2
	Aircraft maximum takeoff weight (MTOW) must not exceed 55 lbs	GR.3
Structures	Aircraft must pass wing tip load test at MTOW	GR.4
	Fuselage must have seperate passenger and cargo compartments	GR.5
	Aircraft must be capable of carrying a minimum of 3 passengers and 1 piece of cargo	GR.6
	Each payload compartment must have a rigidly attached door/hatch	GR.7
Propulsion	Propulsion system and propeller must be commercial off-the-shelf (COTS) parts	GR.8
	Total propulsion energy ≤ 100 Wh	GR.9
	Separate receiver/servo battery required	GR.10
	Arming fuse is a blade-style commercial fuse and does not exceed 100 A or the battery discharge	GR.11
	Fuse holder must be at least 6 in from any propellor	GR.12
	Arming fuse must be mounted on the top surface, fully exposed and safely accessible	GR.13
Manufacturing	All components must be secured, and permanent fasteners must use secondary retention	GR.14
Avionics	Aircraft must have an externally accessible switch to turn on the radio control system	GR.15
	Avionics must use COTS radio with fail-safe on legal frequencies	GR.16

Table 3.5: Aircraft Operational Requirements

Category	Requirement	Req. No.
General	The aircraft must enter the staging box in the clean configuration with all payloads removed	O.1
	Aircraft assembly and payload installation must be completed in less than five minutes	O.2
	Aircraft must land structurally intact and come to a complete stop on the paved runway	O.3
M1	Aircraft must complete 3 laps within five-minute window	O.4
M2	Aicraft must complete maximum amount of laps in five-minute window	O.5
	Aircraft must fly with maximum amount of payload possible	O.6
M3	Aicraft must complete maximum amount of laps in five-minute window	O.7
	Aircraft must fly with maximum length banner	O.8
GM	M2 and M3 demonstrations must be completed as fast as possible	O.9

Table 3.6 summarizes the payload-specific requirements mandated by the competition, defining the constraints associated with carrying passengers, transporting cargo, and deploying the banner.

Table 3.6: Aircraft Operational Requirements

Category	Requirement	Req. No.
M2 Passengers	Passengers must be carried in one compartment on a single plane parallel to horizontal	P.1
	Passengers must board the airplane individually (no speed loaders) and be fully restrained in a vertical orientation	P.2
M2 Cargo	Cargo must be carried in a separate cargo bay(s) that may be in front, behind, or below the passengers	P.3
	Cargo must be restrained to prevent movment during all aspects of flight	P.4
M3 Banner	The banner must have a minimum length of 10 in and a maximum aspect ratio (length/height) of 5	P.5
	The banner must have the team's university name or logo on both sides	P.6
	The banner must be stowed externally for takeoff and may not interfere with any flight or control functions	P.7
	The banner must be remotely deployed and released from the aircraft in flight	P.8
	The banner must remain in the vertical orientation during flight	P.9
	The banner must not sustain any damage in flight or on ground impact	P.10

3.3 Translation to Design Requirements

Based on these design requirements, subsystem design goals were developed to guide the conceptual aircraft design, as summarized in Table 3.7. Verification methods are categorized as visual inspection (V), computer simulation (S), or ground/flight testing (T).

Table 3.7: Sub-Team Design Requirements

Sub Team	Requirement	Sub-Team Design Requirement	Verification Method
Aerodynamics + Payload	GR.1-3	Ensure compliance with competition configuration, weight, and WS limits	V
	GR.5-7, P.1, P.3, P.7	Select aircraft configuration suitable for all payloads	V, S
	O.1, O.2, O.6, O.9, P.2, P.4	Verify M2 systems meet restraint, speed, and removal requirements while minimizing weight/volume and optimizing speed	V, T
	O.8, P.5, P.6, P.8-10	Verify banner and mechanism meet competition requirements throughout all aspects of flight while optimizing banner length	V, T
	O.4, O.5, O.7	Minimize drag to acheive target maximum cruise speed	S
	O.6	Maximize lift to hold maximum payload weight	S
Manufacturing\ Avionics	GR.12-16	All aircraft components must be correctly attached and accessible	V
	GR.4, O.3, O.6	Verify the aircraft is structurally sound and capable of supporting all required components and payload while minimizing weight	T
Propulsion	GR.8-11, O.4-7	Optimize flight speed and endurance while verifying full propulsion system meets competition requirements	V, S

3.3.1 Analysis of Design and Mission Requirements

M1: M1 defines the baseline airworthiness requirements for the aircraft, requiring stable, controllable flight and sufficient climb performance. These requirements established initial targets for wing loading, control surface sizing, and power-to-weight ratio to ensure an effective baseline performance envelope.

M2: M2 scoring incentivizes maximizing net income by balancing passenger and cargo capacity, aerodynamic efficiency, and propulsion energy usage. Increasing payload raises total weight and wing loading, while larger wings and higher battery capacity increase drag and operating cost. As a result, M2 drove trade studies to balance payload capability, drag, and energy efficiency.

M3: M3 scoring depends on banner length, laps completed, and RAC, introducing trade-offs between banner-induced drag, WS, and cruise performance. Longer banners increase drag and stability demands, while minimizing WS degrades aerodynamic efficiency. These constraints required careful integration of the banner system to preserve controllability, cruise speed, and stability.

GM: The GM emphasizes rapid payload handling and banner installation, driving design decisions related to subsystem accessibility and layout. These requirements were balanced against structural integrity, weight, and CG constraints.

3.4 Optimization Analysis

Considering the mission-driven design requirements described in Section 3.3, no single solution existed within the design space. WUDBF identified the key design variables requiring optimization based on the competition rules and utilizes a developed in-house MATLAB optimizer to evaluate the design space. The optimizer accepts constraint variables as inputs and outputs dependent variables that maximize the projected competition score.

3.4.1 Scoring Model

A scoring model was implemented to evaluate predicted competition performance. Six independent design variables were included: WS, wing area, cruise velocity, M2 passenger count, M2 cargo count, and propulsion battery capacity. The first three variables influence aerodynamic performance, while the latter three affect competition scoring.

Each variable was limited to twenty discrete values to ensure computational efficiency, forming a six-dimensional design matrix. Mission scores were predicted using scoring equations defined in Section 3.2, with constants estimated from prior WUDBF aircraft and validated using preliminary aerodynamic and propulsion analyses. Feasibility constraints were applied to remove designs violating competition rules or physical limits, including takeoff distance, wing loading, available power, aspect ratio (AR), and fuselage volume. All optimizer inputs, assumptions, and constraints are summarized in Table 3.8.

Predicted mission scores were normalized using the maximum feasible score for each mission. The resulting dataset was used to identify the design variable combination that maximized the total competition score. These optimized values were selected as initial design targets and are summarized in Table 3.9.

Table 3.8: Optimizer Inputs and Assumption

Variable	Value	Description
$C_{d,min}$	0.095	Based on previous WUDBF aircraft testing and preliminary CFD analysis
$C_{L,max}$	1.7	Based on previous WUDBF aircraft testing and preliminary CFD analysis
Maximum power output	2000 W	Depleats a 100 Wh battery within the five-minute flight window
Maximum takeoff length	100 ft	Requires the aircraft to takeoff within the competition limit
Maximum wing loading	2328 lb/ft^2	Derived from an estimated aircraft weight of 22 lb, representing an upper-bound design condition
Wingspan range	3 ft – 5 ft	Selected to capture the full feasible design space permitted by the competition
Minimum aspect ratio	4	Selceted to verify structural feasibility
Wing area range	1 ft^2 – 5.5 ft^2	Selected to bound lift capability and structural feasibility
Battery capacity	75Wh – 100Wh	Bound by minimum energy required to sustain cruise for the full flight window and the competition-imposed maximum
Passenger range	3 – 30	Varied from the competition minimum to the maximum physically achievable within the fuselage
Cargo range	1 – 15	Varied from the competition minimum to the maximum physically achievable within the fuselage
Maximum fuselage volume	200 in^3	Verify payload combination does not exceed expected fuselage volume, based on conceptual design sizing and past WUDBF aircrafts
Cruise velocity range	65 ft/s – 130 ft/s	Varied over a range consistent with values reported by previous DBF teams

To determine the maximum predicted score, the predicted scoring parameters were normalized by the assumed maximum score for each mission. This resulted in a data set of optimized variables to inform initial design targets, which are summarized in Table 3.9.

Table 3.9: Optimizer Results

Parameter	Value
Wing Span (ft)	4.59
Wing Area (ft^2)	5.17
Cruise Speed (ft/s)	105
Passengers (#)	16
Cargo (#)	4
Battery Capacity (Wh)	85

3.4.2 Sensitivity Analysis

A sensitivity analysis was performed to evaluate how variations in key design parameters affected overall performance. The analysis presents the sensitivities of the M2 and M3 scoring metrics, determining the relative influence of each parameter. The results are shown in Figure 3.2.

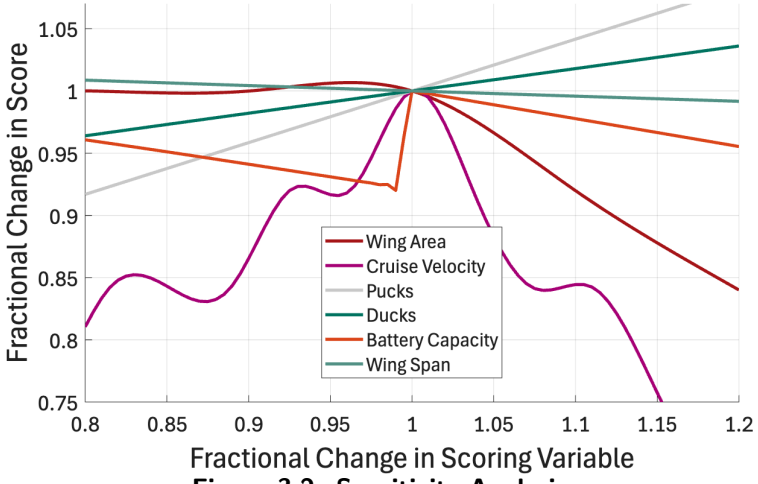


Figure 3.2: Sensitivity Analysis

The sensitivity analysis reveals that increasing the number of ducks and pucks produces a linear increase in score. Among all design variables, passenger count (Ducks) has the strongest positive influence on competition score, followed by cargo count (Pucks). While the optimizer accounts for the added weight of additional payload, it does not model the corresponding increase in aerodynamic drag due to increased fuselage volume or interference effects. Therefore, although payload increases appear highly favorable in the sensitivity plot, real-world performance penalties must be considered during detailed design.

Wing area is the only geometric variable that shows a slight local improvement (a near 3% reduction of baseline area). This improvement is driven primarily by the optimizer's assumption of reduced structural weight with

decreasing wing area. However, this reduction also decreases lift capability and increases wing loading. Given the high-lift requirements for M3 banner towing, where drag is both significant and uncertain, reducing wing area introduces unacceptable risk. Therefore, despite the small numerical gain shown in 3.2, reducing wing area is not operationally desirable.

Wingspan (WS) exhibits an approximately linear trend with a shallow slope. Decreasing WS only slightly increases the score due to its influence on RAC in the M3 scoring equation. However, WS directly affects AR and induced drag, both of which are critical for efficient lift generation. Because lift margin is highly valuable across all missions, particularly for banner towing, WS was not reduced despite the minor scoring sensitivity.

Battery capacity shows a sharp drop in score when reduced below the optimized value, followed and proceeded by a gradual score increase. The steep decline indicates that the optimized capacity corresponds closely to the minimum energy required to achieve an additional lap in each mission. Because laps are discrete integers, the scoring response exhibits step-like behavior: any reduction below the threshold prevents completion of the additional lap, producing a sharp score decrease. Conversely, increasing capacity above the optimized value raises operating cost through the efficiency factor without enabling another lap.

Cruise velocity is the most sensitive design variable. The plot shows a clear peak at the optimized value, with the score decreasing in either direction. Because cruise velocity directly determines lap count within the fixed five-minute window, deviations reduce total mission revenue while weight and propulsion parameters remain constant. This explains the pronounced peak and confirms that cruise speed must be tightly controlled near its optimized value.

3.5 Configuration Selection

To objectively select an optimal aircraft configuration, WUDBF implemented a decision-matrix methodology. This approach evaluates design concepts against weighted performance criteria derived from competition rules. The processes are summarized in Table 3.10.

Table 3.10: Configuration Selection Process

No.	Step	Process
1	Identify Characteristics	Define key design criteria based on rule constraints and mission performance needs
2	Weighting	Assign each characteristic a weight percentage based off relative importance
3	Identify Options	Select feasible configuration concepts based on research and team experience
4	Scoring	Rate each configuration out of 5 (best performance) according to how well it satisfies each criterion
5	Total Score	Multiply scores by weights and sum to determine the highest-performing configuration

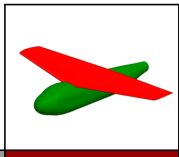
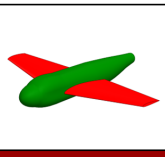
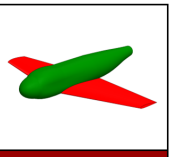
3.5.1 Aircraft Structure Decision Matrices

WUDBF developed four configuration selection matrices to establish the aircraft's overall conceptual layout. The matrices evaluated major architecture-defining components: wing placement, tail configuration, landing gear configuration, and propulsion system.

3.5.1.1 Wing Placement

High-, mid-, and low-wing placements were evaluated in Table 3.11. High-wing designs improve inherent stability but restrict fuselage access and payload bay integration. Mid-wing layouts can be aerodynamically clean, but complicate structure due to a spar through fuselage integration. A low-wing was selected for simpler manufacturing, cleaner aerodynamics, and improved access for passenger loading, cargo bays, and banner mechanism.

Table 3.11: Wing Configuration Decision Matrix

				
Selection Criteria	Weight (%)	High Wing	Mid Wing	Low Wing
Manufacturability	20	3	2	4
Stability	30	5	3	3
Efficency	20	3	5	4
Payload Access	30	3	2	5
	Weighted Score	3.6	2.9	4
	Rank	2	3	1

3.5.1.2 Tail Configuration

Conventional, T-tail, and V-tail configurations were evaluated in Table 3.12. T-tails improve clearance and pitch authority but increase structural complexity, increase drag, and introduce deep-stall risk at low speed. V-tails reduce wetted area but require coupled control mixing and tighter tuning under changing payload and banner loads. A conventional tail was selected based on WUDBF's past aircrafts, low complexity, and sufficient towline clearance without additional reinforcement.

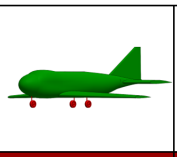
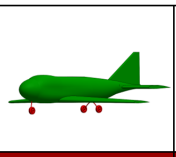
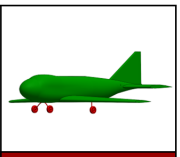
Table 3.12: Tail Decision Matrix

Selection Criteria	Weight (%)	Conventional	T-Tail	V-Tail
Drag	20	5	4	3
Stability	35	4	3	2
Control Complexity	15	5	3	3
Manufacturability	15	5	2	3
Banner Clearance	15	4	5	3
	Weighted Score	4.5	3.35	2.65
	Rank	1	2	3

3.5.1.3 Landing Gear

Tricycle, bow-tricycle, and bow-tricycle-dragger landing gear concepts were evaluated in Table 3.13. A standard tricycle landing gear provides good handling but offers less propeller clearance during rotation and hard landings. The dragger variant improves clearance but can reduce ground stability under heavy loading. The bow-tricycle configuration was selected to provide the best overall balance of stability, handling, propeller clearance, and load support.

Table 3.13: Aircraft Configuration Decision Matrix

				
Selection Criteria	Weight (%)	Tricycle	Bow-Tricycle	Bow-Tricycle Dragger
Takeoff/Landing Stability	30	4	5	3
Ground Handling	20	4	5	3
Propeller Clearence	20	3	5	4
Load Support	20	3	4	4
Weight	10	4	3	3
	Weighted Score	3.6	4.6	3.4
	Rank	2	1	3

3.5.1.4 Propulsion Configuration

Tractor, dual-tractor, and pusher propulsion layouts were evaluated in Table 3.14. Dual-tractor configurations increase thrust potential but increase drag, add weight, and complicate CG management. Pushers can reduce frontal interference but often reduce propulsive efficiency and increase strike risk while tightening CG constraints. A tractor layout was selected for the best balance of efficiency, packaging, and predictable CG relative to payload compartments.

Table 3.14: Propulsion Configuration Decision Matrix

Selection Criteria	Weight (%)	Tractor	Dual Tractor	Pusher
Efficiency	30	4	3	3
Thrust	30	4	5	3
CG Location	20	5	4	1
Drag	20	3	2	4
	Weighted Score	4	3.6	2.8
	Rank	1	2	3

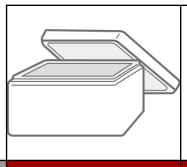
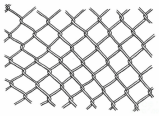
3.5.2 Payload Integration Decision Matrices

WUDBF developed four configuration selection matrices to evaluate payload integration strategies. These matrices assessed factors affecting payload accessibility, internal volume utilization, and mission reliability, including passenger and cargo restraint systems, as well as the banner deployment system and its location.

3.5.2.1 Passenger Restraint System

Foam, lid, and mesh passenger restraint layouts were evaluated in Table 3.15. Foam restraints offer high passanger capacity but add significant weight and manufacturability difficulty, while lid configurations reduce weight but are less space efficient. The mesh configuration was selected for its low weight, high manufacturability, and high space efficiency.

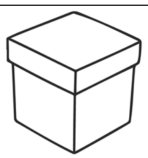
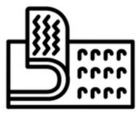
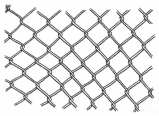
Table 3.15: Passenger Restraint System Decision Matrix

				
Selection Criteria	Weight (%)	Foam	Lid	Mesh
Manufacturability	15	2	3	5
Space Efficiency	30	3	2	4
Weight	15	2	5	4
Passenger Capacity	40	4	3	3
	Weighted Score	3.1	3	3.75
	Rank	2	3	1

3.5.2.2 Cargo Restraint System

Rigid lid, Velcro, and mesh net cargo restraint concepts were evaluated in Table 3.16. The mesh configuration offered advantages in manufacturability, space efficiency, and weight, but failed to provide adequate cargo capacity, which was the highest weigth criteria. The Velcro constraint significantly reduced weight, but failed to maximize cargo capacity. The lid restraint system was selected for cargo because although heavier, it demonstrated superior cargo capacity and high manufacturability.

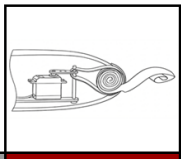
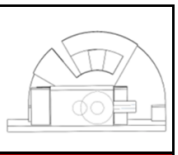
Table 3.16: Cargo Restraint System Decision Matrix

				
Selection Criteria	Weight (%)	Lid	Velcro	Mesh
Manufacturability	15	4	3	4
Space Efficiency	30	2	2	4
Weight	15	3	5	4
Cargo Capacity	40	5	3	2
	Weighted Score	3.65	3	3.2
	Rank	1	3	2

3.5.2.3 Banner Deployment Mechanism

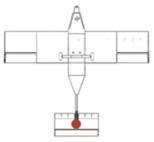
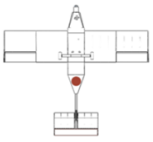
Rotary plate, quick-release pin, and servo–Velcro release concepts were evaluated in Table 3.17. The servo–Velcro system was rejected due to inadequate reliability and manufacturability. Although the quick-release pin offered a lightweight solution, it did not meet reliability requirements. The rotary plate was ultimately selected, as it provided the most reliable option while maintaining low weight.

Table 3.17: Banner Deployment Mechanism Decision Matrix

				
Selection Criteria	Weight (%)	Servo + Velcro	Quick Release Pin	Rotary Plate
Reliability	50	2	3	5
Weight	20	4	5	5
Manufacturability	30	3	5	4
	Weighted Score	2.7	4	4.7
	Rank	2	3	1

3.5.2.4 Banner Mechanism Location

Table 3.18: Banner Deployment Mechanism Location Decision Matrix

				
Selection Criteria	Weight (%)	Tail	Rear Fuselage	Mid Fuselage
CG Stability	40	2	4	5
Landing Clearance	20	3	5	5
Banner Clearence	40	5	4	2
	Weighted Score	3.4	4.2	3.8
	Rank	3	1	2

Three banner mechanism locations were evaluated in Table 3.18: beneath the wing spar, at the rear fuselage, and on the tail. A tail-mounted configuration was rejected due to poor center-of-gravity control and large induced moments. An under-wing spar location improved mass centralization; however, it introduced a risk of banner interference with the aircraft structure. The rear fuselage location minimized drag and induced moments and was selected despite modest CG-balancing challenges.

3.6 Final Conceptual Design

The final conceptual design has a WS of 4.59 ft and a wing area of 5.17 ft^2. The aircraft is configured to carry 16 passengers secured by a mesh locking restraint system, along with 4 cargo items restrained within a conventional box-style cargo bay. The target cruise speed is 105 ft/s and the target battery capacity is 85 Wh. The design is a low-wing aircraft featuring a conventional tail and a tractor-configured propulsion system with a bow-tricycle landing gear arrangement. A rotary plate banner release mechanism is integrated at the aft end of the fuselage.

4 Preliminary Design

The initial aircraft configuration developed in the conceptual design phase was refined during the preliminary design phase. A combination of constraint sizing and aerodynamic and propulsion analysis were used to arrive at a final preliminary design. The specific methods used are outlined in the following section.

4.1 Design Methodology

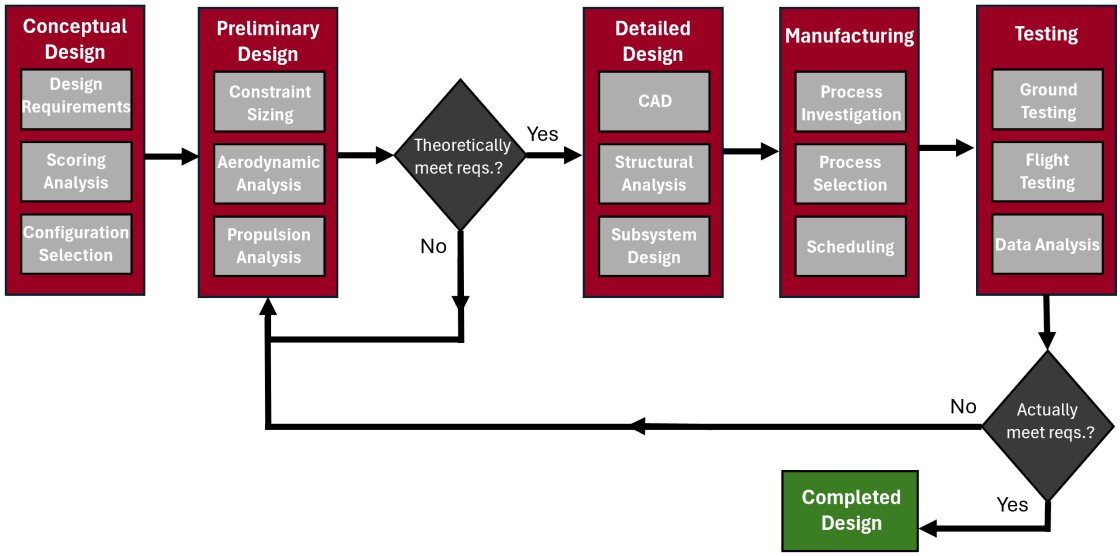
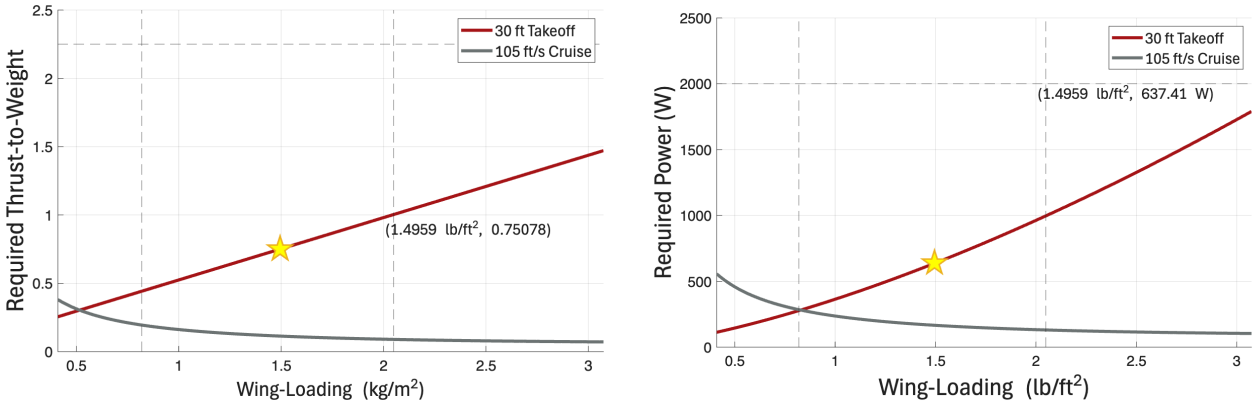


Figure 4.1: Design Methodology

The aircraft design followed a structured, iterative methodology, detailed in Figure 4.1, progressing from conceptual through preliminary, detailed design, manufacturing, and testing. Design reviews were conducted following the preliminary design and testing phases to evaluate performance, requirement compliance, and technical risk. These reviews served as decision points to determine whether the design could advance or required refinement in earlier phases, ensuring the final aircraft met mission requirements while balancing performance, manufacturability, and reliability.

4.2 Constraint Sizing

An initial constraint sizing was performed by plotting the required thrust-to-weight ratio (T/W) to achieve a 30 ft takeoff and 105 ft/s cruise speed for a range of wing loadings using the equations developed in Gudmundsson [2]. WUDBF chose a 30 ft takeoff limit to verify that the plane will takeoff in a reasonable time. Since the aircraft must be propeller driven using an electric motor, it is useful to convert T/W to required electrical power using initial estimates of weight, mechanical efficiency, and propulsive efficiency. This meant that the upper power constraint of 2000 W listed in Table 3.8 could be applied in the constraint sizing. The assumed maximum T/W is 2.25 based on previous WUDBF aircrafts. The constraint sizing graphs can be found in Figure 4.2.



(a) T/W vs Wing Loading **(b) Power vs Wing Loading**

Figure 4.2: Constraint Sizing

Subfigure 4.2b illustrates that the upper power constraint of required power and T/W is obeyed by all wing loading values. Wing loading of 1.50 lb/ft^2 was calculated using the initial M1 (empty aircraft) weight estimation detailed in Section 4.3 and wing area detailed in Section 3.2. From the chosen wing-loading of 1.50 lb/ft^2, the minimum required power for take-off was set at about 637.41 W and the minimum required T/W ratio at about 0.75, given by Figure 4.2.

4.3 Initial Weight Estimation

The all up weight (AUW) of the aircraft was estimated using a breakdown of individual components. The weight of the wing, horizontal tail (HT), vertical tail (VT), and fuselage were estimated using previous aircrafts, which were built with similar methods and materials. Battery, motor, propeller, electronics, carbon fiber (CF) rods, and landing gear weights were determined by weighing representative components already possessed by WUDBF. The M2 mass quantity was determined using the optimization program described in Section 3.4. The M3 banner mechanism mass was estimated using SolidWorks, and the banner mass was estimated using an approximation of the prospective material's density, and added weights for stability.

Table 4.1: Initial Weight Estimation

Component	Weight (lb)
Propulsion Battery	1.38
Motor + Propellor	1.00
Fuselage	2.85
CF Tail Boom	1.60
CF Wing Spar	1.30
Electronics	0.33
Avionics Battery	0.32
Wing	0.40
HT	0.08
VT	0.05
Rear Landing Gear	0.20
Front Landing Gear	0.10
Banner Mechanism	0.30
Total M1 (Empty)	**9.91**
Passengers	0.64
Cargo	1.48
Total M2	**12.03**
Banner	1.00
Total M3	**10.91**

4.4 Aerodynamic Performance

The aerodynamic design of the aircraft was iteratively improved to ensure that it is stable, controllable, and achieves maximum aerodynamic performance. XFLR5 [3], MATLAB [4], SolidWorks Flow [5], and Ansys [6] were the primary tools used for aircraft aerodynamic analysis.

4.4.1 General Sizing

The wing geometry selected during the conceptual design phase resulted in a WS of 4.59 ft and a wing area of 5.17 ft^2. At the estimated MTOW, this corresponds to a wing loading of 2.33 lb/ft^2. This sizing provides sufficient lift capability for M2 payload configurations while maintaining acceptable stall margins and limiting drag growth for M3. The selected geometry served as the baseline for the aerodynamic and stability analyses presented in the following sections.

4.4.2 Airfoil Selection

To identify potential airfoil candidates, data was analyzed from multiple online repositories, including Airfoil Tools [7] and the UIUC Airfoil Coordinates Database [8]. Performance characterization was conducted using XFLR5, which employs a linear vortex panel method coupled with an integral boundary layer formulation, iteratively solving for pressure distributions to accurately predict viscous effects and separation points at low Reynolds numbers. In order to select the main wing airfoil, the predicted Reynolds number (Re) of the aircraft was first calculated using Equation 4.1.

$$Re = \frac{MAC \cdot V_\infty}{\nu} \tag{4.1}$$

where MAC is the mean aerodynamic chord (ft), V_∞ is the freestream velocity (ft/s), and ν is the kinematic viscosity of air (ft/s^2). The kinematic viscosity was determined from the average temperature in April(23 °C) [9] and the altitude(1300 ft) [10] of Wichita, Kansas which resulted in a kinematic viscosity of $\nu = 1.7 \cdot 10^{-4} ft^2/s$. The MAC and freestream velocity were taken from the optimizer, described in section 3.4, which resulted in a

Reynolds number of $Re = 580,000$.

Candidate airfoils were evaluated based on their lift-to-drag ratio (L/D) at a target cruise lift coefficient of $C_L = 0.2$, representative of nominal cruise conditions. Airfoil performance was evaluated at a representative Re corresponding to cruise flight. Geometric constraints were applied to ensure structural feasibility, including a minimum thickness requirement to accommodate the primary wing spar. In addition, a minimum required maximum lift coefficient ($C_{L,max}$) was imposed to ensure adequate takeoff and low-speed performance. Five airfoils were selected as finalists and are summarized in Table 4.2. Figures 4.3 and 4.4 compare the lift characteristics and aerodynamic efficiency of the selected airfoils, with α representing the angle of attack in the C_L vs α plot.

Table 4.2: Analyzed Airfoils

Airfoil	Max Cl/Cd	Max. Thickness [% of chord]	Max. Camber [% of chord]
MH-84	94.84	13.7	4.1
NACA 2412	89.35	12	2
NACA 4412	109.31	12	4
NACA 4418	93.81	18	4
NACA 5412	114.07	12	5

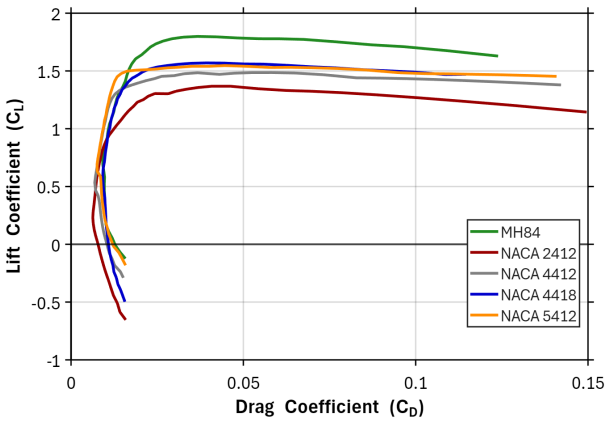


Figure 4.3: C_D vs. C_L Drag Polar

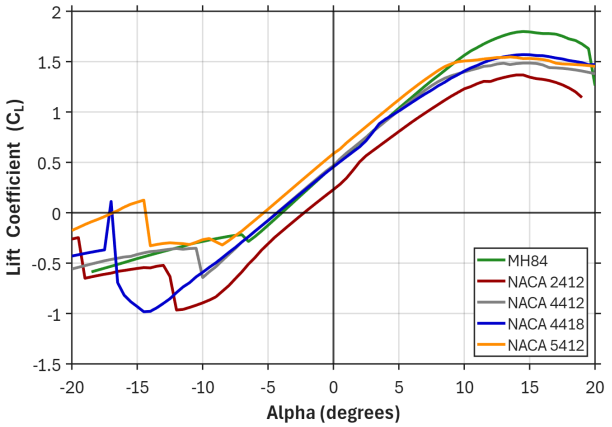


Figure 4.4: C_L vs. α Comparison

4.4.3 Empennage Sizing

An aircraft with competitive maneuverability and nominal drag was desired to minimize lap times and maximize battery endurance. Additionally, the empennage design factored in the banner mechanism location to ensure no interference with banner towing. WUDBF selected the NACA 0010 airfoil as it is common for stabilizers and used successfully in past WUDBF projects [2]. Initial sizing of the empennage for stable flight began by investigating the vertical tail volume (V_{VT}) and horizontal tail volume (V_{HT}) for aircraft of the same class. For a single-engine aircraft, the initial estimate was set at around $V_{HT} = 0.70$ and $V_{VT} = 0.04$ [2]. Initial horizontal and vertical tail areas were then solved for using Equation 4.2 and 4.3 [2]:

$$S_{HT} = \frac{V_{HT} \cdot S_{ref} \cdot c_{ref}}{l_T} \tag{4.2}$$

$$S_{VT} = \frac{V_{VT} \cdot S_{ref} \cdot b_{ref}}{l_T} \tag{4.3}$$

Where S_{HT} is the horizontal tail area (ft²), V_{HT} is the horizontal tail volume, S_{VT} is the vertical tail area (ft²), V_{VT} is the vertical tail volume, S_{ref} is the main wing area (ft²), c_{ref} is the geometric chord (ft), b_{ref} is the span of the main wing (ft), and l_T is the optimal tail arm length (ft). Based on these calculations, the VT and HT planform areas were set at 1.72 ft² and 3.33 ft² respectively. An initial assumption was made that the AR of the HT would be equal to about half the AR of the main wing and that the AR of the VT would be about 2.2 [2].

Initial estimates for the span of the VT and HT were then made using Equation 4.4 and 4.5 as shown below [2]:

$$b_{HT} = \sqrt{AR_{HT} \cdot S_{HT}} \tag{4.4}$$

$$b_{VT} = \sqrt{AR_{VT} \cdot S_{VT}} \tag{4.5}$$

Where b_{HT} is the horizontal tail span (ft), and b_{VT} is the vertical tail span (ft). Based on XFLR5 simulations, a 0.5 taper ratio was selected for the vertical stabilizer to minimize induced drag, whereas the horizontal stabilizer required no taper. The preliminary tail dimensions are detailed in Table 4.3

Table 4.3: Preliminary Tail Dimensions

	Area (ft^2)	Wing Span (ft)	MAC (ft)	Taper Ratio
VT	0.65	1.64	0.75	0.5
HT	1.29	1.57	1	-

4.4.4 Control Surface Sizing

Given the traditional design of this aircraft, all control surfaces (rudder, elevator, and ailerons) were designed using rule-of-thumb parameters from Gudmundsson [2]. Control surface sizing was verified using Equation 4.6 (see below) from Gudmundsson [2], previous WUDBF aircraft, and prototype performance.

$$C_{l_{\delta_\alpha}} = \frac{c_{l_{\delta_\alpha}} c_R}{Ab}\left[(b_2^2 - b_1^2) + \frac{4(\lambda - 1)}{3b}(b_2^3 - b_1^3)\right] \tag{4.6}$$

Where $C_{l_{\delta_\alpha}}$ is the aileron authority, $c_{l_{\delta_\alpha}}$ is the change in lift coefficient with aileron deflection, b is the wing span (ft), A is the wing area (ft²), c_R is the root chord (ft), and λ is the taper ratio.

The hinge point of the ailerons was placed at the center of the leading edge arc to minimize large gaps and overhangs when the surfaces are deployed. The ailerons have a chord fraction of 30% and a span of 50% of the wing, designed to deploy between $\pm$ 30°. Chord fractions for the control surfaces on the empennage were 30% for the elevator, and 40% for the rudder. Control surfaces on both the rudder and elevator cover 100% of the arm and were designed to deflect a maximum of 30°. The summarized control surface data can be found in Table 4.4. Using data taken from XFLR5, Fig. 4.5 shows the plotted lift coefficient versus α with the ailerons and elevator deployed at 30° and with no control surfaces activated.

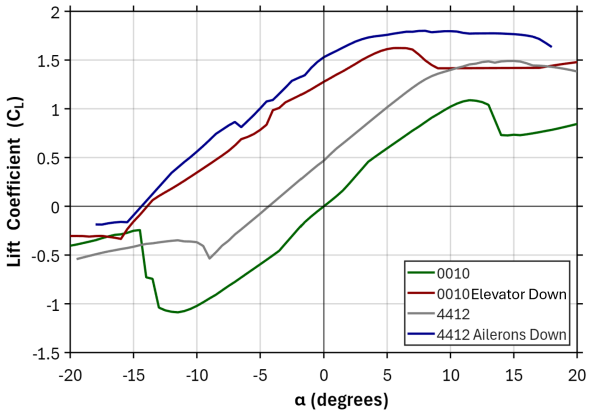


Figure 4.5: C_L vs. α for Various Airfoil Configurations

Table 4.4: Preliminary Control Surface Dimensions

	Chord %	Span %
Aileron	30	50
Elevator	30	100
Rudder	40	100

4.4.5 Landing Gear

Table 4.5: Preliminary Landing Gear Dimensions

Parameter	Value
Rear Landing Gear	
Length (in)	12.5
Width (in)	0.98
Height (in)	5.51
Front Landing Gear	
Height (in)	6.88

Landing gear geometry was designed to provide adequate propeller and banner release clearance while maintaining stable ground handling and minimizing parasitic drag. Geometric constraints were evaluated using the aircraft CAD model to ensure sufficient clearance during takeoff rotation and banner release operations.

A target takeoff rotation angle of 8° was selected to balance lift generation with propeller ground clearance, based on wing incidence, landing gear height, and tail clearance considerations. This rotation angle was achieved by designing the nose landing gear to be taller than the main landing gear. To ensure ground stability,

the main landing gear was oriented perpendicular to the fuselage centerline to provide resistance to roll during taxi, takeoff, and landing. The preliminary landing gear dimension can be found in Table 4.5.

4.4.6 Aircraft Lift and Drag Analysis

A simplified model of the aircraft was analyzed using ANSYS and SolidWorks Flow simulations at the optimal cruise airspeed to estimate the aerodynamic drag force acting on the aircraft. The resulting drag force was then used to calculate the aircraft parasitic drag coefficient using Equation 4.7.

$$C_{d,min} = \frac{2F_D}{\rho u^2 A} \tag{4.7}$$

Where F_D is the simulated drag force (lbf), ρ is air density (slug/ft^3), u is cruise velocity (ft/s^2) and A is the wing area (ft^2). This analysis yielded a total parasitic drag coefficient of $C_{d,min} = 0.05$. At cruise conditions, the total combined drag force is projected to be 3.31 lbf, with parasitic effects contributing 2.49 lbf. As shown in Figure 4.9, the fuselage and wings are the primary contributor to the aircraft's parasitic drag. While the fuselage generated higher drag than anticipated, the penalty is offset by the increased area for payload storage.

Finally, the parasitic drag coefficient obtained from CFD was superimposed onto the lift-induced drag calculated using XFLR5, where the induced drag coefficient was found to be $C_{d,i} = 0.01$. The combined contributions were used to generate the total aircraft drag presented in Figure 4.7.

Lift was also evaluated using computational fluid dynamics (CFD) under the same cruise-condition simulations. The simulated lift force was used to compute the lift coefficient according to Eq. 4.8, where F_L is the simulated lift force (lbf).

$$C_L = \frac{2F_L}{\rho u^2 A} \tag{4.8}$$

At cruise conditions, the aircraft generated a total lift force of 28.75 lbf, corresponding to a lift coefficient of $C_L = 0.20$. Based on the predicted lift and drag values, the projected lift-to-drag ratio of the aircraft at cruise is 8.7.

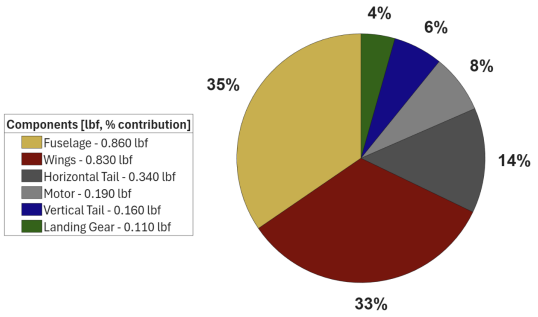


Figure 4.6: Parasitic Drag by Component

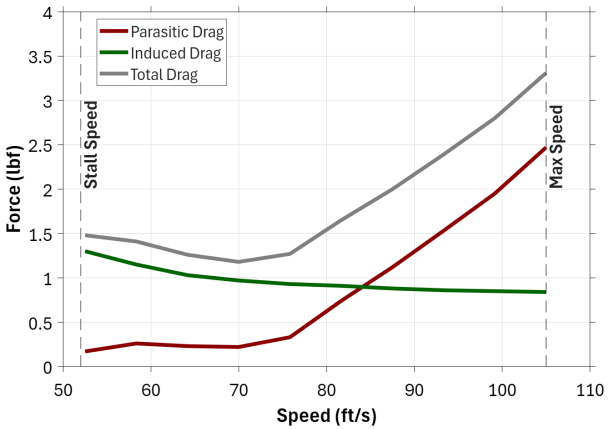


Figure 4.7: Lateral Mode Roots.

4.4.7 Banner Drag Estimation

During M3, a banner payload is deployed and towed behind the aircraft. In steady, level flight, the banner is designed to remain approximately aligned with the freestream, such that the resulting aerodynamic drag is dominated by skin-friction effects. Additional drag contributions arise from minor misalignment, fabric deformation,

and low-amplitude flutter.

To ensure acceptable aircraft performance during banner towing, a minimum lift-to-drag ratio of 5:1 was established as a design requirement for M3 flight. Based on the predicted aircraft lift at cruise, this constraint corresponds to a maximum allowable banner drag of 2.4 lbf; exceeding this value would reduce the overall lift-to-drag ratio below the acceptable threshold.

To support banner sizing and propulsion analysis, a parametric banner drag model was developed to relate tow drag to banner geometry, material properties, and flight conditions. This model was used to evaluate candidate banner areas and AR and to identify the banner size at which drag penalties would outweigh scoring benefits. The final banner configuration selected using this analysis is defined in Section 4.6.3 and is referenced when computing the effective M3 tow drag.

4.4.7.1 Banner Drag Model

The aerodynamic drag generated by the banner is modeled as the sum of pressure drag and skin-friction drag, as shown in Eq. 4.9.

$$D_t = D_p + D_f \tag{4.9}$$

Where D_t is the total banner drag (lbf), D_p is the pressure drag (lbf), and D_f is the friction drag (lbf).

The pressure drag component is given by Equation 4.10[11].

$$D_p = \frac{1}{2}\rho u^2 C_{d,p} A \tag{4.10}$$

Where ρ is air density (slug/ft^3), u is airspeed (ft/s), $C_{d,p}$ is the drag coefficient, and A is the area of the banner (ft^2). The pressure drag coefficient was taken as $C_{d,p} = 0.19$, based on flag data reported by Munsons [11]. Skin-friction drag was computed using Eq. 4.11. Since friction acts on both sides of the banner, the effective wetted area is twice the planform area.

$$D_f = \rho u^2 C_{d,f} A \tag{4.11}$$

The banner material was assumed to be 13-oz vinyl, resulting in a conservative effective skin-friction coefficient of $C_{d,f} = 0.01$. For the banner drag estimation analysis, the cruise speed was taken from the optimizer results presented in Section 3.4. Figure **??** presents the predicted banner drag as a function of banner area, with a constraint line at 2.4 lbf corresponding to the minimum allowable lift-to-drag ratio of 5:1 during M3 flight. As shown, banner areas exceeding this limit would generate excessive drag and degrade aircraft performance. Based on this analysis, the maximum allowable banner area was determined to be 0.92 ft^2. A pie chart of parasitic drag by component for M3 is shown in 4.10, with the total parasitic drag being 4.89 lbf, and total drag force being 5.81 lbf.

4.5 CFD Trade Study

As part of the aerodynamic refinement process, additional aerodynamic devices were investigated to improve overall aircraft efficiency and mission performance. In particular, reducing induced drag was identified as a key objective due to its direct impact on cruise velocity, lap time, and energy consumption across multiple missions. Three candidate wingtip configurations were considered: a planar endplate, a triangular winglet, and an arrowhead-

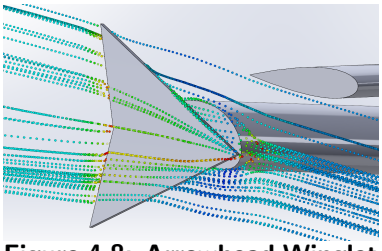


Figure 4.8: Arrowhead Winglet CFD

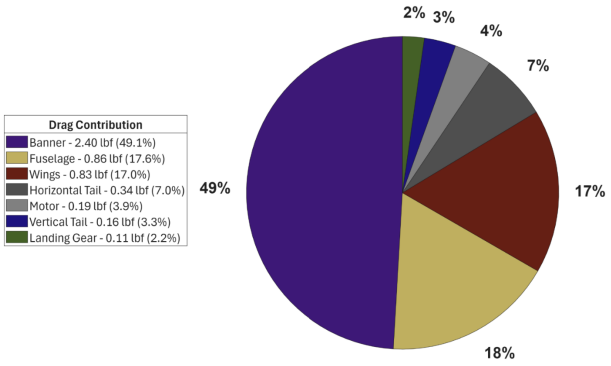


Figure 4.9: Parasitic Drag by Component

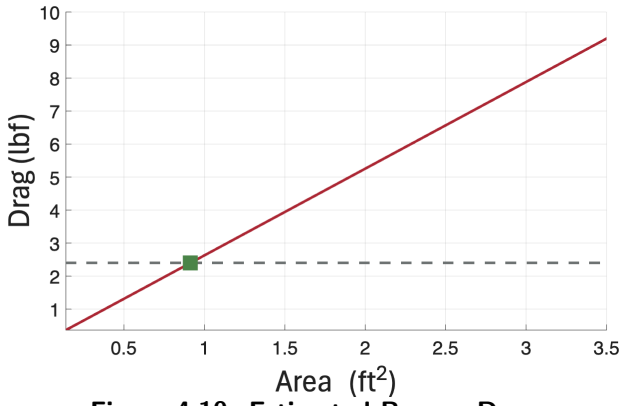
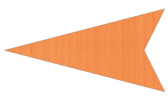


Figure 4.10: Estimated Banner Drag

style winglet. CFD simulations were conducted at the predicted mission cruise speed of 105 ft/s for each wing configuration. Figure 4.8 shows the streamlines near the wingtips for the arrowhead concept, with the color scale representing the strength of rotational flow. Reduced vortex intensity corresponds to lower induced drag. Using the CFD-derived drag results, a decision matrix was constructed to evaluate the three wingtip configurations based on aerodynamic performance, added mass, and stability impact, shown in Table 5.4.

Table 4.6: Wingtip Decision Matrix

				
Selection Criteria	Weight (%)	Triangle	Arrowhead	Quarter-Elipse
Drag	30	4	3	3
Weight	30	3	5	4
Stability	40	3	5	4
	Weighted Score	3.3	4.4	3.7
	Rank	2	1	3

Based on the decision matrix results, the arrowhead-style winglet was selected for the final aircraft design. This configuration provided the most favorable combination of drag reduction, minimal weight penalty, and acceptable stability characteristics. The selected arrowhead winglet increases spiral mode stability by 20% relative to the clean wing. The final winglet geometry consists of a max height of 8 in, a sweep angle of 14.5°, a length of 13.54 in, and a thickness of 0.125 in, shown in Figure 4.11

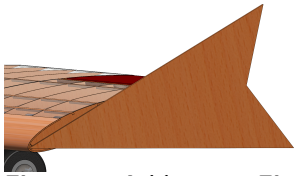


Figure 4.11: Final Winglet Design

4.6 Stability Analysis

4.6.1 Static Stability

The entire aircraft geometry was simulated in XFLR5 [3]. A detailed CG prediction was established using four key inputs: the expected payload mass, component weights for the motor and electronics, historical weight averages for the wing and empennage, and a preliminary fuselage design. To calculate the static margin (SM) of the aircraft, Equation 4.12 was used.

$$SM = \frac{X_{NP} - X_{CG}}{MAC_{Wing}} \times 100\% \tag{4.12}$$

Where SM is the static margin (%), X_{NP} is the neutral point of the aircraft in the x (longitudinal) direction [ft], X_{CG} is the CG of the aircraft in the x direction [ft], and MAC_{Wing} is the mean aerodynamic chord of the wings [ft]. X_{NP}, X_{CG}, and MAC_{Wing} were automatically calculated by XFLR5.

At cruise conditions, the SM was calculated to be 11% for M1, 14% for M2, and 12% for M3. Figure 4.12 illustrates the variation of pitching moment coefficient with angle-of-attack for each mission configuration.

The slope of the pitching moment curves, which is proportional to SM, remains relatively constant across the analyzed angle-of-attack range. This behavior indicates consistent longitudinal stability and positive static stability for all mission configurations

4.6.2 Dynamic Stability Analysis

XFLR5 was used to evaluate the aircraft's dynamic stability characteristics at cruise conditions. The resulting root-locus plot, which represents the system eigenvalues in the complex plane, was analyzed to assess the stability of the primary longitudinal and lateral–directional modes. Initial analysis indicated weak stability in the spiral and phugoid modes.

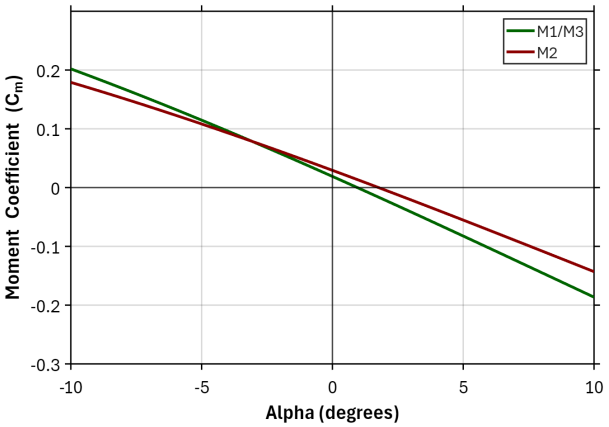


Figure 4.12: C_M vs. α for Different Missions

To improve lateral–directional stability, a 4° dihedral was incorporated into the wing design. The final root-locus plots, shown in Figure 4.13 and Figure 4.14, demonstrate stable behavior in the roll, Dutch roll, spiral, and short-period pitch modes, as evidenced by eigenvalues with negative real components.

The phugoid mode exhibited marginal stability, with eigenvalues located close to the imaginary axis. This behavior indicates lightly damped, low-frequency oscillations that evolve slowly relative to pilot response time and are therefore considered acceptable for safe and controllable flight. With these changes, the stability characteristics for all missions achieved Level 1 flying qualities as assessed by MIL-F-B-785C. A summary of the aircraft's dynamic eigenmode characteristics is provided in Tables 4.7 and 4.8.

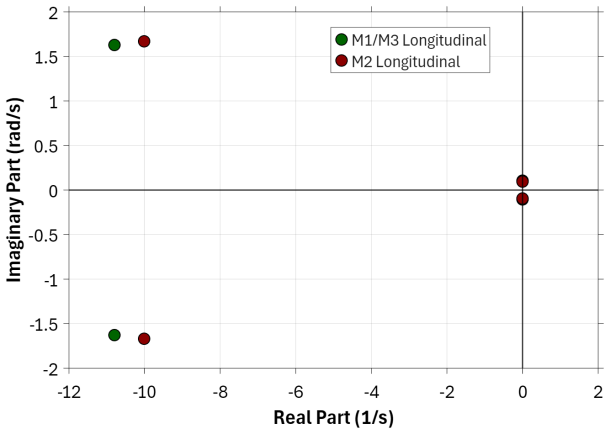


Figure 4.13: Longitudinal Mode Roots.

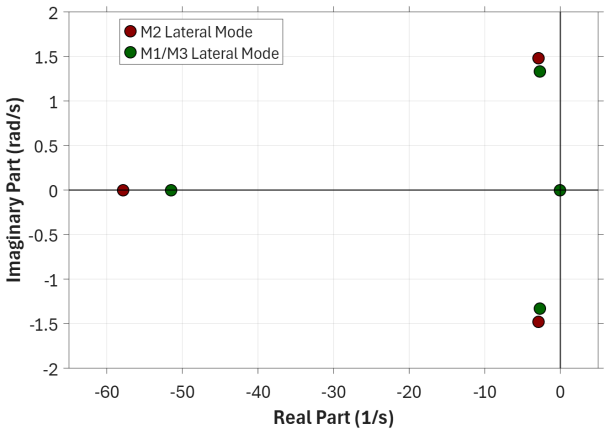


Figure 4.14: Lateral Mode Roots.

Table 4.7: M1/M3 Dynamic Stability Values

Mode	Damping Ratio ζ	Natural Frequency ω_n (rad/s)	Damped Natural Frequency ω_d (rad/s)	Time Period τ (s)
Short Period	0.726 (0.35<ζ<2.0)	2.37	1.63	-
Phugoid	0.008 (ζ>0.004)	0.103	0.103	-
Roll	-	-	-	0.013 (τ<1.4)
Dutch Roll	0.306 (ζ>0.17)	1.397 (ω_n>1.0)	1.33 (ω_d>0.35)	-
Spiral	-	-	-	11.47

Table 4.8: M2 Dynamic Stability Values

Mode	Damping Ratio ζ	Natural Frequency ω_n (rad/s)	Damped Natural Frequency ω_d (rad/s)	Time Period τ (s)
Short Period	0.691 (0.35<ζ<2.0)	2.31	1.67	-
Phugoid	0.0106 (ζ>0.004)	0.0955	0.0955	-
Roll	-	-	-	0.012 (τ<1.4)
Dutch Roll	0.299 (ζ>0.17)	1.59 (ω_n>1.0)	1.48 (ω_d>0.35)	-
Spiral	-	-	-	16.69

4.7 Mission Payloads

4.7.1 M2 Passengers

The passenger payload consists of standardized rubber ducks provided by the competition, which exhibit small variations in size. An open-configuration passenger compartment was designed to maximize volumetric efficiency and loading flexibility. Unlike partitioned layouts, the absence of fixed seating allows accommodation of the maximum number of passengers regardless of individual size variations.

Figure 4.15: Preliminary Passenger Restraint System

The compartment base was sized to accommodate at least 16 passengers arranged in two rows of eight, extending along the fuselage. This corresponds to a base footprint of approximately 5 in by 20 in, bounded by a 1 in tall perimeter wall for external support, as shown in Figure 4.15. The structure was manufactured using 3D-printed polylactic acid (PLA), selected for its low weight and manufacturability.

To secure the passenger payload, a flexible fabric mesh was implemented to keep the ducks in a vertical orientation during all phases of flight. The mesh conforms to the passengers' profiles, allowing it to accommodate variations in duck height while preventing in-flight movement.

4.7.2 M2 Cargo

The cargo payload is housed in a single compartment designed to secure standard hockey pucks. Located beneath the passenger compartment, the cargo container features a duct-tape–hinged lid to enable rapid loading and unloading during the GM. The compartment was sized to hold 4 pucks arranged in a 2×2 configuration, resulting in a square shape with 6.25 in sides. The structure was manufactured using 3D-printed PLA, selected for its low weight and manufacturability. The full integration of both passenger and cargo restraint systems are shown in Figure 4.16.

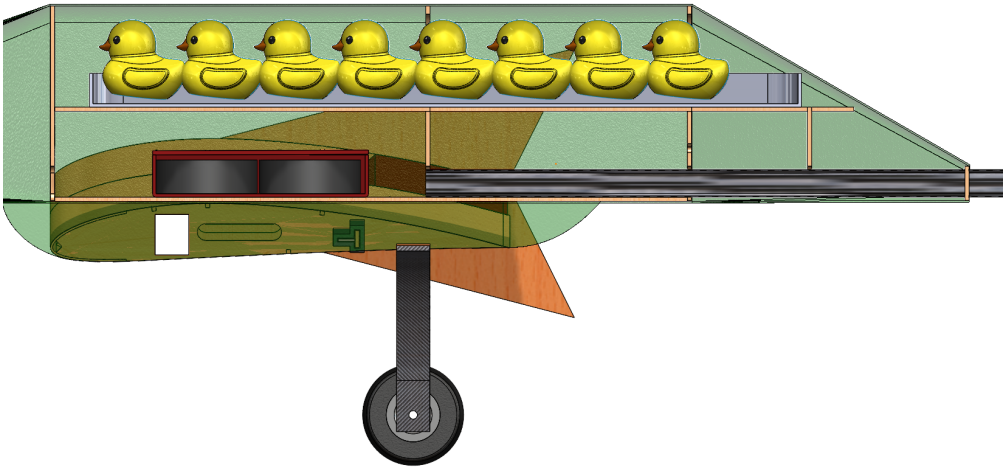


Figure 4.16: Final M2 Configuration

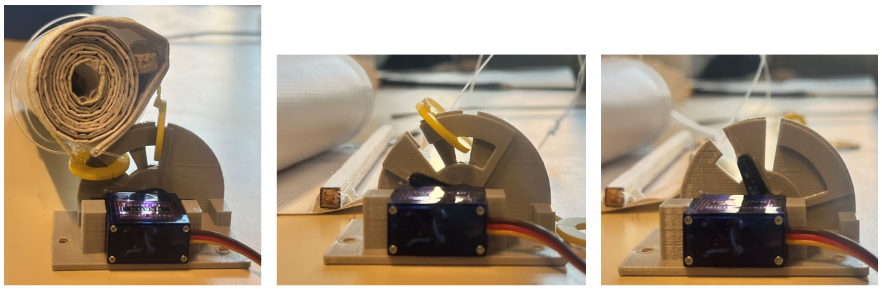
4.7.3 M3 Deployment Mechanism

To minimize system weight and complexity, the team prioritized a unified mechanism capable of performing three distinct functions: retention (stowing), deployment (unfurling), and jettison (release). Early trade studies evaluated linear pin-pullers and drop-away shell concepts; however, these were rejected due to their relatively poor reliability and manufacturability. A rotary latch mechanism was ultimately selected for its high functional density, allowing all three operational states to be controlled by a single actuator. The preliminary design features a concentric rotary plate assembly driven by a servo. The mechanism consists of a stationary stator plate and a rotating rotor plate, both featuring specific slot geometries, shown in Figure 4.17. The banner's tow line, and rubber band retention guard, are held in these slots. As the servo rotates the rotor plate, the alignment of the slots changes, transitioning the system through its operational stages:

Stage I: Retention (Stowed State): The servo is held at the neutral position ($0°$). The rotor plate physically blocks the exit path of the tow line (attached the rightmost washer), and rubber band retention guard (attached to both washers), keeping the banner folded and secured against the fuselage.

Stage II: Deployment (Towing State): The servo actuates to the intermediate position (approx. $30°$). This rotation aligns the retention slot, allowing the rubber band guard (left washer) to detach from the mechanism, letting the banner bundle to fall away from the aircraft, but stay attached with the tow line connected to the right washer.

Stage III: Jettison (Release State): To jettison the banner, the servo rotates to its maximum travel limit (around $70°$). This fully aligns the release slot with the right washer, allowing the banner to completely separate from the aircraft. The drag force and gravity acting on the banner free the tow line from the mechanism, cleanly separating the banner from the aircraft.



Stage I Stage II Stage III

Figure 4.17: Mechanism Stages

4.7.4 M3 Banner

The competition rules give a maximum banner AR of 5:1 (length-to-height). While classical fluid mechanics theory [11] indicates that increasing AR generally increases both pressure drag and skin-friction drag, the mission scoring framework prioritizes banner area rather than aerodynamic efficiency. Because the Mission 3 scoring equation scales linearly with banner area, selecting the maximum allowable AR provides the greatest potential scoring benefit. As discussed in Section 4.4.7, a parametric analysis identified an optimal banner area of 0.92 ft^2, corresponding to a banner 5.15 in in height and 25.74 in in length.

A 13 oz vinyl (polyvinyl chloride) banner material was selected over lighter alternatives such as ripstop nylon, Tyvek, or coated paper. Although vinyl exhibits greater surface roughness and consequently higher skin-friction drag, its increased stiffness provides improved aerodynamic stability and prevents tearing, which would violate the competition rules. The added stiffness promotes consistent deployment behavior, reduces susceptibility to folding or tangling during flight, and ensures repeatable performance across multiple mission attempts.

The banner rigging system utilizes 0.03 in diameter high-strength fishing line arranged in a V-shaped tow configuration. The tow lines are attached to grommets along the leading edge of the banner to distribute loads evenly and improve lateral stability. A carbon fiber dowel attached to the leading edge further enhances stiffness and helps maintain a consistent banner orientation during towing.

4.8 Propulsion

The propulsion system was designed using the outputs of the performance optimizer presented in Section 3.4, which identified a target propulsion battery capacity of 85 Wh and a nominal cruise speed of 105 ft/s. Unlike previous competition years, total propulsion battery capacity directly influences the M2 EF and therefore the overall mission score, as shown in Equation 3.7. Separately, the constraint sizing analysis established minimum T/W of 0.75 to ensure acceptable takeoff and climb performance at the maximum takeoff weight, as detailed in Section 4.2. Additional considerations included battery availability at the optimized battery capacity, endurance, and maximizing efficiency across all three missions.

4.8.1 Battery Selection

The AIAA DBF rules permit the use of lithium-polymer (LiPo) or nickel-metal hydride batteries for propulsion. LiPo batteries were selected due to their higher energy density, discharge rate, and electrical efficiency, which are well suited for the high power demands of the propulsion system.

For an aircraft of this size and weight, LiPos below 6-cell (6s) are unlikely to provide the performance required by the optimizer. For 10-cell (10s) LiPos, they are not readily available below the 100 Wh (2777.78 mAh for 10s) limit. Therefore, an 8-cell (8s) LiPo configuration was evaluated relative to a 6s configuration. The higher nominal voltage of an 8s battery (29.6 V) compared to a 6s (22.2 V) allows the propulsion system to deliver equivalent power at lower current, as given by Equation 4.13.

$$P = V * I \tag{4.13}$$

Reducing current draw decreases potential resistive losses and thermal loading within the propulsion system, improving overall efficiency. This is explained by Joule's first law, which states that the heat generated in a system is proportional to the square of the current, the resistance of the wire, and the time the current flows. The decrease in heat generation leads to increased efficiency as less energy is turning to heat and more is going to power the system. Additionally, for a given motor and propeller, the increased voltage enables higher achievable

rotational speed and thrust margin, as motor no-load speed scales linearly with voltage per Equation 4.14

$$RPM = Kv * V \tag{4.14}$$

Results from the optimizer indicated that a battery capacity near 85 Wh provided a favorable balance between scoring penalties associated with energy capacity and the power required to support high-load flight segments. Based on these results and component availability, a MaxAmps 8s 82.88 Wh (2800 mAh) LiPo battery with a 25C-Constant and 100C-Burst discharge rate was selected for the propulsion system.

4.8.2 Motor and Propeller Selection

Candidate motor-propeller combinations were evaluated using the online propulsion analysis tool eCalc [12]. The search was constrained using the performance requirements summarized in Table 4.9.

Table 4.9: Ecalc Input Variables

Parameter	Value	Reasoning
Battery	8s, 82Wh	Detailed in section 4.8.1
T/W	0.751	Detailed in section 4.2
Max Cruise Velocity (ft/s)	105	Detailed in section 3.4
Max Propeller Diameter (in)	17	Estimated Max Ground Clearance
Endurance (min)	5	Maximum flight window

An initial set of over 50 candidate motor–propeller combinations satisfied the imposed constraints. Configurations were filtered based on compliance with these requirements, component availability, cost, and the inclusion of safety factor in both flight time and cruise speed to account for off-nominal operating conditions. This process reduced the candidate set to the three motor-propeller combinations listed in Table 4.10.

Table 4.10: Motor-Propeller Combinations

Motor	Propeller	Max Velocity (ft/s)	RPM	T/W	Static Thrust (lbf)	Max Current (A)	Total Mass (lb)
ScorpionA-5025-310	15x10	128	8378	1.11	12.05	41.80	1.15
Scorpion SII-4035-330	15x10	132	8677	1.15	12.52	49.50	0.96
SunnySky V3 x4130 V3 380Kv	15x10	146	9575	1.39	15.12	63.30	0.86

A two-blade, 15 in diameter, 10 in pitch propeller was selected, as maximizing diameter within geometric constraints increases thrust, while the chosen pitch provides a balance between static thrust and cruise efficiency across the operating envelope for all three motors. This also allows the motor–propeller trade study to focus on differences in motor performance.

All shortlisted configurations met or closely approached the stated performance requirements in Sections 3.4 and 4.2. To further differentiate between candidates, comparative performance metrics were generated using eCalc, including electrical efficiency and endurance as a function of speed, static thrust as a function of static electrical power, and speed as a function of throttle setting.

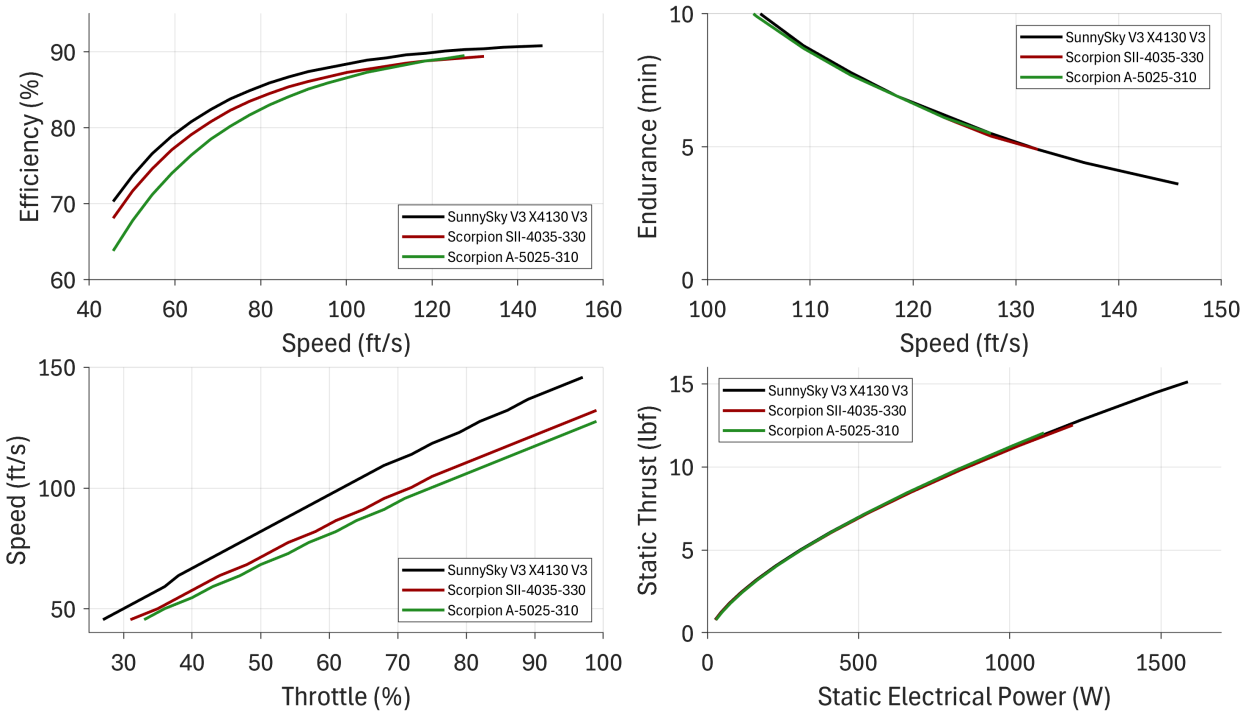


Figure 4.18: Motor Performance Comparisons

Based on these comparisons, the SunnySky V3 X4130 V3 380 Kv motor paired with a 15×10 propeller demonstrated the highest electrical efficiency at a relevant cruise velocity range while providing favorable available thrust and access to higher speeds relative to the other candidates.

4.8.3 ESC and Fuse Selection

Based on the predicted maximum current draw of approximately 63.30 A for the selected motor–propeller–battery configuration, the Phoenix Edge Lite 8s 75 A electronic speed controller (ESC) was selected. The ESC provides adequate continuous current capability and is compatible with the system's 8s LiPo battery voltage. A 70 A blade fuse was selected to match the battery pack's maximum continuous discharge current while accommodating nominal and transient peak loads.

4.8.4 Final Propulsion System Details

Table 4.11: Final Propulsion System Details

Battery	Motor	Propeller	ESC	Fuse
MaxAmps LiPo 2800mAh 8s 29.6V Battery Pack	SunnySky X Series V3 X4130 V3 Brushless Motors 380 kv	APC 15x10 E	PHOENIX EDGE LITE 75 A ESC, 8S / 33.6V WITH 5 A BEC	70 A Maxi Blade Fuse

4.8.5 Required vs. Available Thrust Analysis

The dynamic performance of the propulsion system was evaluated to verify eCalc-predicted results and determine the aircraft speed envelope. As eCalc does not directly provide dynamic thrust data, propeller performance data from the APC Propellers database [13] was used. This dataset is generated using vortex theory with propeller geometry modeled through the NASA Transonic Airfoil Analysis Computer Program [14].

Predicted rotational speeds from eCalc were used to query the APC dataset, and corresponding dynamic thrust values were obtained using an interpolation script developed in MATLAB. Required and available thrust curves are shown in Figure 4.19.

Based on this analysis, the estimated maximum aircraft velocity was 121 ft/s. Differences between this value and eCalc-predicted speeds are attributed to simplified drag modeling assumptions within eCalc, whereas the present analysis more explicitly accounts for aircraft aerodynamic drag, resulting in a refined estimate of achievable cruise performance.

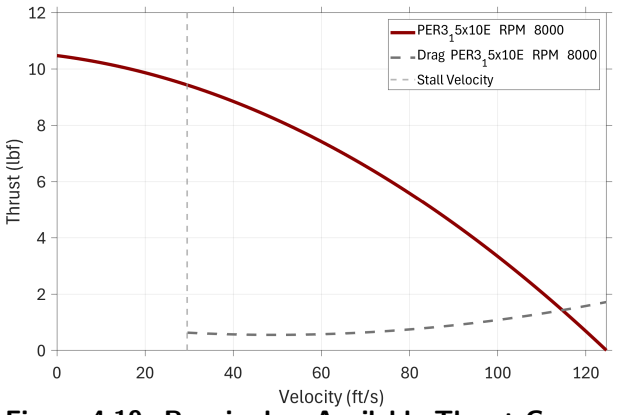


Figure 4.19: Required vs Available Thrust Curves

4.9 Predicted Aircraft Performance

In order to analyze the various aspects of the preliminary design and predict overall aircraft performance, WUDBF utilized Km_sim, an in-house kinematic mission simulator written in MATLAB [15].

4.9.1 Brief Explanation of Km_sim

Km_sim works by simulating an aircraft's performance as it flies through a mission profile. Each mission profile consists of a series of functions based on kinematics and Newton's second law describing the motion of the aircraft during a particular phase of flight (e.g., takeoff, climb, cruise, etc.). Specific mission profiles were written for M1, M2, and M3. Aerodynamic and propulsion data from Sections 4.4 and 4.8, respectively, were used to generate aerodynamic and propulsion performance look-up tables. These tables allow the simulation to determine the lift, drag, and thrust of the aircraft at any particular point in time. During a simulation, Km_sim steps through each mission phase, numerically integrating the relevant equations of motion using ode45 and the look-up tables, while keeping track of changes in the state variables. For a complete explanation of Km_sim and the source code see [15].

Table 4.12: Preliminary Km sim Results

	M1	M2	M3
Inputs			
All-Up Weight (lb)	9.91	12.03	10.91
Wing Loading (lb/ft^2)	1.92	2.33	2.11
$C_{L,max}$	1.67	1.67	1.67
Parasitic Drag (lbf)	2.49	2.49	4.89
Cruise Throttle (%)	70	70	70
Outputs			
Takeoff Distance (ft)	23.2	29.1	28.9
$C_{L,cruise}$	0.12	0.15	0.21
$C_{D,cruise}$	0.03	0.03	0.09
Cruise Velocity (ft/s)	115	114	92
Number of Laps	3	8	6
Time at End of Final Lap (min)	1.81	4.82	4.65

4.9.2 Km_sim Results

Separate simulations for M1, M2, and M3 were carried out in Km_sim to estimate performance in competition. The results of this analysis are shown in Table 4.12.

4.10 Uncertainties

The design tools and analysis methods used during the preliminary design process inherently possessed uncertainty. Aerodynamic lift and induced drag were computed in XFLR using the vortex lattice method (VLM), which neglects lifting surface thickness, air viscosity, and fuselage effects. Based on previous WUDBF designs, the reduction in lift due to neglected viscosity was approximately offset by the exclusion of fuselage lift, and wing thickness drag was small relative to total drag. As a result, the VLM-predicted lift and induced drag were considered representative of the full aircraft.

Parasitic drag estimates using Gudmundsson's method also contained uncertainty, which was partially mitigated through CFD analysis. Propulsion performance was estimated using eCalc and the APC propeller database; although these tools are third-party and employ proprietary models, their accuracy has been demonstrated in prior WUDBF aircraft. Mission performance was evaluated using Km_sim, which assumes ideal pilot inputs and neglects wind effects, and therefore includes uncertainty effects from aerodynamic and propulsion models. Overall, uncertainties were managed by comparing outputs from multiple analysis methods and refining model inputs until consistent performance predictions were obtained.

5 Detail Design

5.1 Dimensional Parameters

Building on the outcomes of the preliminary design, the detailed design phase focused on refining the aircraft's structural components, payload systems, and external geometry to meet performance, manufacturability, and mission requirements. Particular emphasis was placed on reducing structural weight, streamlining component integration, and minimizing the GM time. The resulting key dimensions of the finalized configuration are summarized in Table 5.1.

Table 5.1: Final Dimensional Parameters

Wing		Horizontal Stabilizer	
Airfoil	NACA 4412	Airfoil	NACA 0010
Span (ft)	4.59	Span (ft)	1.57
Chord (ft)	1.12	Chord (ft)	0.820
Dihedral (°)	4	AR	1.92
AR	4.06	**Verticle Stabilizer**	
Fuselage		Airfoil	NACA 0010
Length (ft)	2.93	Root Chord (ft)	0.984
Width (ft)	0.583	Tip Chord (ft)	0.492
Height (ft)	0.482	Taper Ratio	0.5
Propulsion System		**Overall Aircraft**	
Propeller	15×10	Length (ft)	4.70
Battery	82.88 Wh 8s LiPo	Width (ft)	4.59
Motor	SunnySky V3 X4130 V3 380 Kv	Height (ft)	1.67
ESC	Phoenix Edge Lite 75A	Empty Weight (lbs)	7.68
M2 Payload		Max Weight (lbs)	10.88
Passengers (#)	16	**Avionics System**	
Cargo (#)	4	Receiver	FrSky X8R Receiver
Banner		Battery	Tattu 11.1V 75C 3S 850mAh Lipo Battery Pack
Length (in)	75	Transmitter	FrSky 2.4GHz Taranis Q X7

5.2 Structural Design Goals and Methodology

WUDBF's primary structural design goals were to maximize strength and stability, improve accessibility to electronics and payload compartments, minimize weight, and simplify the manufacturing process. The overall structure is designed to direct loads from each subsystem into the major load-bearing components of the aircraft. The fuselage consists of a fiberglass monocoque with a CF boom running from the fuselage's center-firewall to the tail. Propulsive loads from motor thrust and torque that can cause bending, torsion, and vibrations are directed through a motor mount and distributed into the fuselage. Ground loads from the landing gear are directed through the fuselage structure and into the central CF boom. These load paths are illustrated in Fig. 5.1. The CF spars

interface with an aluminum bracket and span the full wing, providing bending and torsional stiffness. Aerodynamic loads from the wing are transferred to the wing box, where they are then distributed across the fuselage structure and the central CF rod. The empennage is directly connected to the central CF rod via a CF spar and structural attachment, distributing loads into the central boom and fuselage.

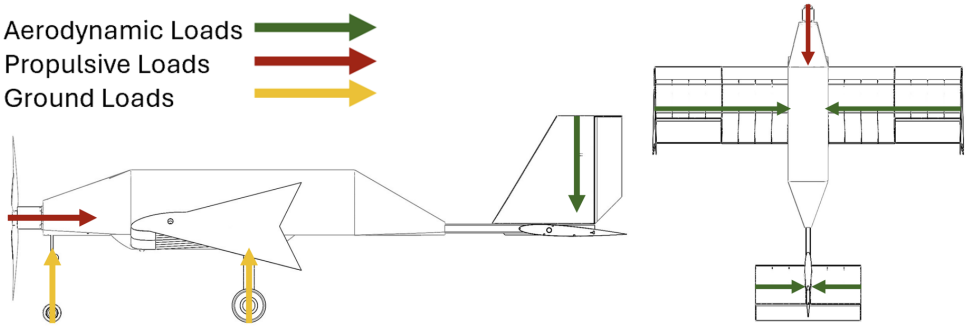


Figure 5.1: Load path diagram

5.3 Subsystem Design

5.3.1 Fuselage Design

The fuselage is a fiberglass composite shell reinforced with internal plywood firewalls, providing high structural rigidity. The configuration is tailored to accommodate the volume of the M2 passenger and cargo payloads, and avionics and propulsion components, with the internal geometry detailed in Figure 5.3. Each compartment is accessed through its own fiberglass hatch with a duct-tape hinge. A tapered, rectangular fuselage geometry was selected to maximize volume and flat floor area.

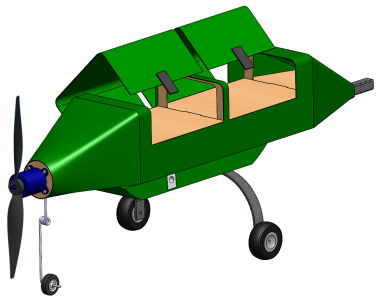


Figure 5.2: Detailed Fuselage Assembly

A carbon fiber tail boom is bonded to the aft fuselage and empennage. A belly fairing is mounted beneath the fuselage to house the wing dihedral mount and provide rear landing gear attachment points. The belly fairing features tapered leading and trailing edges, with thickness concentrated around the dihedral stand to minimize parasitic drag and pressure buildup, and is bolted to the fuselage at four attachment points to constrain motion in all directions.

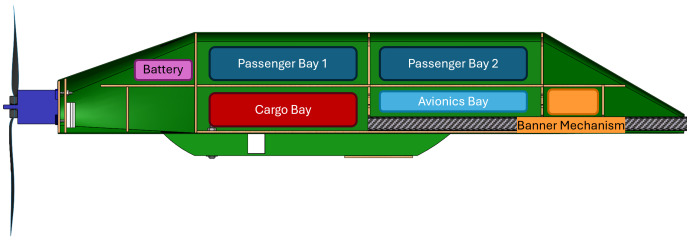


Figure 5.3: Detailed Fuselage Internal Bays

5.3.2 Wing Design

The wing structure consists of 9 ribs per wing, with a single CF spar running through each wing panel. The inboard end of each CF spar interfaces with the dihedral mount, which sets the wings at 4° above the horizontal plane. The dihedral mount is fixed in a rectangular cutout within the belly fairing beneath the fuselage, positioning the stand flush with the upper surface to ensure accurate alignment and efficient load transfer.

To resist wing rotation and bending loads at the root, the innermost rib of each wing incorporates structural tabs that are attached to the fuselage sidewalls. Servo mounts and control surface attachment points are integrated directly into the wing structure to ensure precise control alignment.

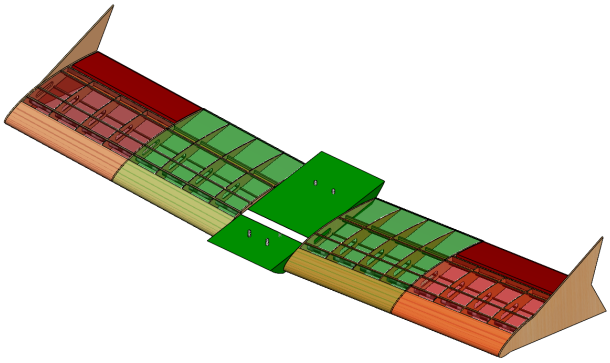


Figure 5.4: Detailed Wing Assembly

The control surfaces are fabricated from foam cores with a fiberglass layup to provide adequate stiffness and resistance to damage. The wing control surfaces are controlled by servos via pushrods, transmitting pilot inputs to the center of the control surfaces. The leading edges of the wing ribs are skinned with balsa to increase local and torsional stiffness, improving load distribution along the span. Finally, Monokote is heat-shrunk over the wings to provide a smooth aerodynamic surface finish while minimizing additional weight.

The wing assembly, showing ribs, CF rod, leading edge, servos, and control surfaces, is shown in Fig. 5.4.

5.3.3 Empennage Design

The empennage was designed using the same structural philosophy as the main wing to ensure stiffness, low mass, and manufacturing consistency. The horizontal stabilizer is supported by a single CF rod running through the entire span, providing the primary bending stiffness. The vertical stabilizer does not incorporate a CF rod; instead, a plywood tab extends through its lower ribs and interfaces with the horizontal stabilizer CF rod, creating a rigid mechanical connection between the two surfaces.

Both the horizontal and vertical stabilizers are bonded to the carbon fiber tail boom extending from the fuselage. The rudder control surface is constructed similarly to the wing control surfaces, using reinforced foam. However, the elevator control surface is fabricated from tapered balsa wood, which has higher stiffness than the reinforced foam. The empennage control surfaces are also controlled by servos via pushrods, transmitting pilot inputs to the center of the control surfaces. Similar to the wing, servo mounts and hinge attachment points for the rudder and elevator, are integrated directly into the rib structure to ensure precise control alignment. The completed empennage structure is shown in Fig. 5.5.

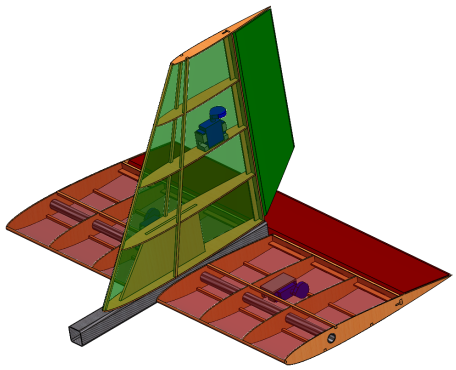


Figure 5.5: Detailed Empennage Assembly

5.3.4 Landing Gear

The landing gear system was implemented using COTS components, as WUDBF did not possess the equipment required to manufacture carbon fiber landing gear in-house. COTS components were selected based on experience from previously WUDBF aircrafts. A JTEC RADIOWAVE JTC-E10 bow-style tricycle rear landing gear was selected for the rear configuration due to its geometric compatibility with the aircraft design and its durability, as shown in Fig. 5.6. A COTS aluminum alloy tricycle front landing gear assembly with an integrated steering wheel was selected and modified by adjusting its height to achieve the desired take-off angle. The front landing gear is retained using the pin-mount system supplied with the kit, which constrains translational motion while allowing free rotation.

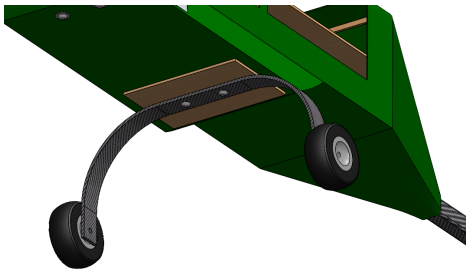


Figure 5.6: Detailed Rear Landing Gear

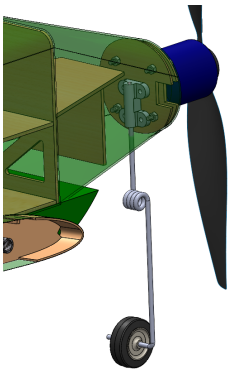


Figure 5.7: Detailed Front Landing Gear

Steering control is provided by a servo connected to the top of the front landing gear via a pushrod, transmitting pilot inputs to the wheel. This configuration enables controlled taxiing, takeoff, and landing operations, as shown in Fig. 5.7.

5.3.5 M2 Payloads

The passenger restraint system is rigidly attached to the wooden fuselage floor, providing a fixed and repeatable passenger location that supports consistent center-of-gravity placement. During loading, the ducks are placed individually into the container, after which the mesh restraint system is installed to secure the payload. The mesh is fastened using snap fasteners spaced evenly around its perimeter, with the female fasteners bonded to the mesh and the male fasteners bonded to the container walls. Additionally, the interior base of the container is lined with double-sided tape to increase stablity.

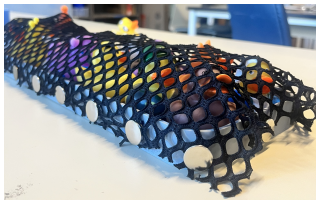


Figure 5.8: Detailed Passenger Restraint

The cargo payload, described in Section 4.6.2, contains the pucks, which are packed in direct contact to minimize the overall fuselage cross-section. The cargo canister is fully removable from the aircraft, allowing the pucks to be loaded and unloaded outside of the fuselage to reduce handling time. Once loaded, the canister is secured within the cargo bay.

Located directly beneath the passenger deck, the cargo bay is accessed through a secondary, smaller hinged door. Due to vertical height constraints within the lower fuselage and the instability associated with an edge-standing orientation, the pucks are stowed horizontally. The cargo restraint mechanism, which ensures positive retention during all phases of flight. A side view showing the how the full passenger and cargo configuration will look inside the plane is shown in Figure 4.16.

5.3.6 M3 Payloads

5.3.6.1 Banner Mechanism

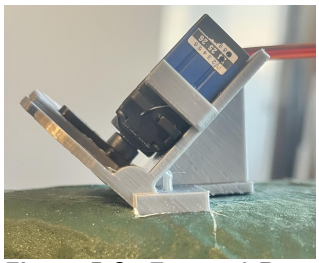


Figure 5.9: External Banner Release Mechanism

The banner deployment and jettison mechanism is actuated by a single Spektrum A332 servo controlled through an onboard microcontroller. Pilot commands are transmitted from a Traxxas TQ transmitter to a Spektrum SR315 receiver, which outputs standard PWM control signals. These signals are routed to an Arduino Uno, which interprets the incoming channel commands and maps them to discrete servo positions corresponding to different stages of the mechanism operation. The Arduino rewrites the receiver PWM signal to command the A332 servo arm to predetermined angular positions rather than directly passing through pilot input.

The mechanism is built with a U shaped bracket that will be attached around the CF tail boom inside the aft end of the fuselage, and will remain attached for all missions. The external geometry of the mechanism was altered to allow the tow line to fall at a 30° angle off the fuselage, to restrict the possible contact with the aircraft structure, as seen in Figure 5.9.

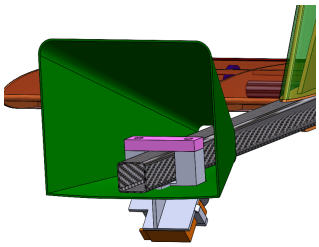


Figure 5.10: Internal Banner Release Mechanism

5.3.6.2 Banner

After conducting wind tunnel testing, detailed in Section 7.3, the banner dimensions were decided to be 12 in tall by 60 in long. During flight an added mass at the bottom of the leading edge dowel was decided to be 0.20 lbs, made of washers tied

around a fishing line loop, as seen in Figure 5.11. Four separate 4.50 in diameter, semi circular slits were cut at 60% length and 80% length. The slits are made to help with visibility and decrease drag across the banner. The WashU logo will be hand painted onto the banner prior to competition.

5.4 Final Weights and Locations

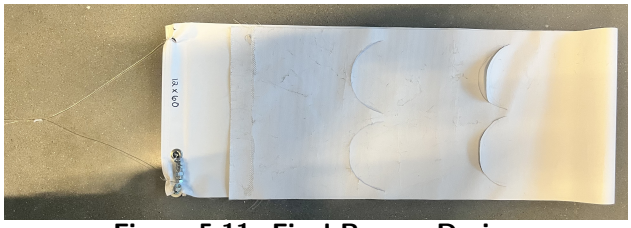


Figure 5.11: Final Banner Design

Each mission results in a unique mass configuration, so a consistent method was required to evaluate the aircraft's CG. All component masses and locations were extracted from the final CAD assembly to enable a full weight-and-balance analysis. The coordinate system was defined with its origin at the leading edge of the central wing rib: the positive X-axis extends aft of the origin towards the tail, the positive Y-axis extends out the right wing (starboard) when viewed from behind, and the positive Z-axis points upward. The masses and absolute coordinates of all major aircraft components are listed in Table 5.2. Mission-specific CG locations for M2 and M3 were computed by adding the corresponding passenger and cargo payloads. Minor mass items, such as bolts, adhesives, covering, and wiring, are excluded from the table but were incorporated into the global CG estimate through distributed allowances in the CAD model. The final M1 configuration naturally places the CG within the acceptable range, so no additional ballast was required.

Table 5.2: Final Weight Balance

Component	Mass (lb)	X-position (in)	Y-position (in)	Z-position (in)
Wing	1.13	1.97	0	0.35
Fuselage	2.83	2.97	0	2.7
VT	0.09	43.31	0	2.3
HT	0.14	41.34	0	0.67
Rear Landing Gear	0.21	13.78	0	-0.3
Front Landing Gear (inc. servo)	0.13	-5.2	0	-0.5
4 Control Surface Servos	0.18	2.36	0	0.4
Avionics Battery	0.24	6.3	0	2.8
Propulsion Battery	1.41	-5.91	0	2.8
Motor + Propeller	0.99	-11.81	0	0.25
Banner Mechanism	0.33	19.69	0	-0.23
M1 \ Empty				
Overall	7.68	1.43	0	1.70
M2				
Passengers + Restraint Systen	0.44	10.4	0	3.1
Cargo + Restraint System	2.76	3.95	0	0.2
Overall	10.88	2.85	0	1.88
M3				
Banner	0.4	19.69	0	-2.95
Overall	8.08	2.45	0	1.56

5.5 Structural Performance

In order to verify structural performance of the aircraft, WUDBF utilized Solidworks Finite Element Analysis (FEA) [16] to simulate the major structural components of the aircraft under their maximum expected load conditions.

5.5.1 Motor Forces

Motor-induced loads represent a significant source of stress on the fuselage, particularly under static and low-speed operating conditions such as take-off. Static thrust and torque values were obtained from the propulsion analysis summarized in Table 4.10 and from propulsion bench testing. The motor mount and fuselage were designed to withstand a maximum static thrust of 15.12 lbf, corresponding to the worst-case take-off condition.

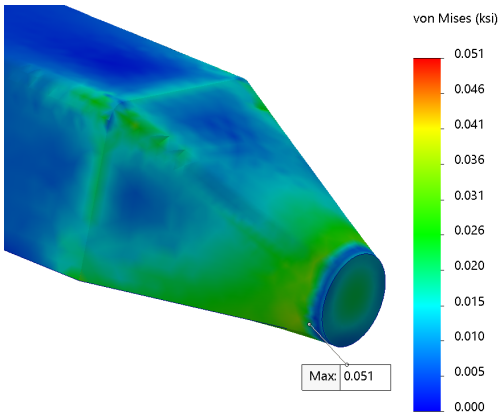


Figure 5.12: Fuselage and Motor Mount Von Mises Stress

The resulting design load was applied to the motor mount and forward fuselage structure in the SolidWorks FEA. The resulting von Mises stress distribution on the fuselage and motor mount is shown in Figure 5.12. The analysis provided a maximum body von Mises stress of 51 psi, located at the lower region near the interface between the motor mount and the fuselage. This stress level remains below the yield strength of both the fiberglass shell and the birch plywood mount, verifying the component will remain within safe limits during all expected conditions.

5.5.2 Wing Loading Force

FEA was performed in SolidWorks on the CF wing spar to verify that the wing structure could withstand the high lift loads associated with the M2 payload configuration. The spanwise lift distribution was approximated using XFLR5[3] for the design cruise condition. To apply this distributed aerodynamic loading within SolidWorks, the discrete lift data from XFLR5 was curve-fit in MATLAB to generate a continuous spanwise lift distribution, as shown in Fig. 5.13. The lift distribution is presented as a fraction of the total lift as a function of normalized wing span. Using the fitted lift distribution, loading was applied to the wing model and constrained at the interface between the aluminum wing bracket and the CF spar to simulate the transfer of aerodynamic loads into the fuselage through the dihedral mount.

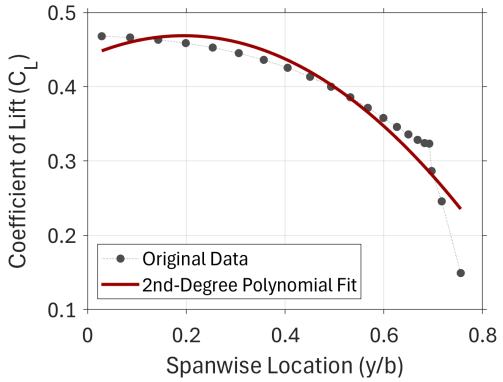


Figure 5.13: Lift Force Across Wing

The analysis showed peak stress concentrations at the aluminum wing bracket–spar interface, where bending loads are highest, where the maximum von Mises stress was predicted to be 8 ksi. Carbon fiber composite does not exhibit yielding behavior; therefore, failure was assessed relative to the material ultimate strength. A conservative allowable stress of 50 ksi was used [17], corresponding to a safety factor of 6.26. Under the lift force, the maximum vertical deflection occurred at the wingtip and was calculated to be 0.20 in.

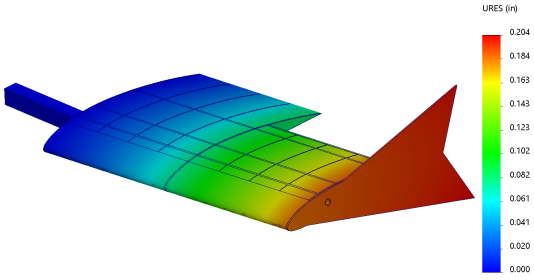


Figure 5.14: Wing Displacement

5.5.3 Tail Loading Force

5.5.3.1 Horizontal Stabilizer

FEA was performed in SolidWorks on the CF HT spar to verify structural integrity under the high lift loads associated with the M2 payload configuration. The spanwise lift distribution at the design cruise condition was obtained from XFLR5 [3] with the elevator at maximum downward deflection to represent a conservative loading case. This distribution was applied to the model as a continuous distributed load following curve fitting of the discrete aerodynamic data, as shown in Figure 5.15. The lift distribution is expressed as a fraction of total lift along the normalized WS.

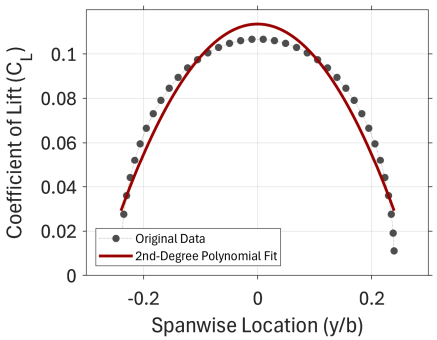


Figure 5.15: Lift Force Across Horizontal Stabilizer

The wing was constrained at the intersection of the HT spar and the CF tail boom to represent load transfer into the fuselage. Peak stresses were observed at this interface, where bending loads are highest, as seen in Figure 5.16. The maximum von Mises stress was predicted to be 7 ksi, creating a safety factor of 7.35. The maximum vertical deflection occurred at the wingtip and was found to be 0.03 in, which is sufficiently small that it is not expected to result in material failure or permanent deformation.

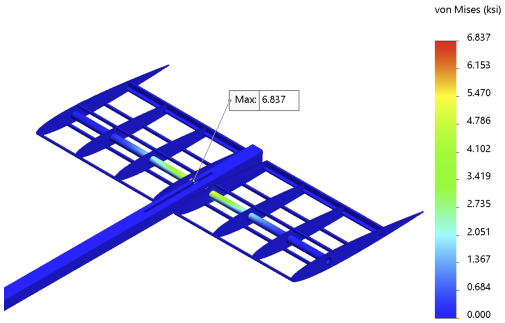


Figure 5.16: Horizontal Stabilizer Von Mises Stress

5.5.3.2 Vertical Stabilizer

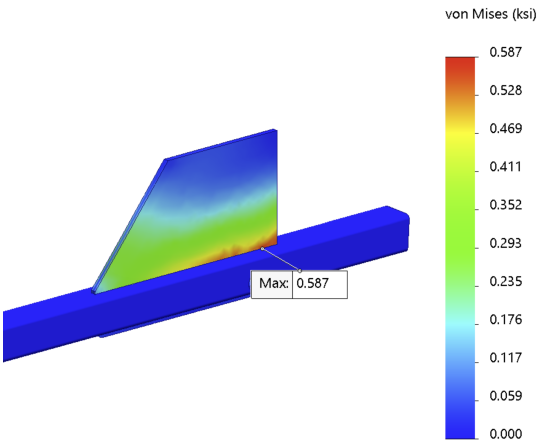


Figure 5.17: Vertical Stabilizer Von Mises Stress

The same analysis procedure was applied to the wooden stabilizer within the vertical stabilizer, which is constructed from birch wood, as shown in Figure 5.5. The aerodynamic loading was calculated with the rudder at maximum deflection to represent a conservative loading case. The stabilizer was constrained at the interface between the wooden member and the carbon fiber tail boom to represent load transfer into the fuselage. Peak stresses were observed at this interface, where bending loads are highest, as seen in Figure 5.17. The maximum von Mises stress was predicted to be 587 psi, corresponding to a safety factor of 12.30. The maximum horizontal deflection occurred at the stabilizer tip and was calculated to be 0.05 in, which is sufficiently small that it is not expected to result in material failure or permanent deformation.

5.5.4 Rear Landing Gear Loading FEA

FEA was performed in SolidWorks on the rear landing gear assembly to verify structural integrity under a conservative landing load case. To represent a harsh landing scenario, a vertical load equal to three times the maximum aircraft weight of M2 was applied to the landing gear, equating to 32.64 lbf.

The CF rear landing gear was constrained at the

interface between the landing gear mount and the fuselage structure to represent load transfer into the airframe. Peak stress concentrations were observed at the mounting interface and at the wheel mounts, as seen in Figure 5.18. The maximum von Mises stress was predicted to be 15 ksi, creating a safety factor of 3.36. Under the applied load, the maximum vertical deflection of the rear landing gear was calculated to be 0.03 in, which will have little to no effects on the CF.

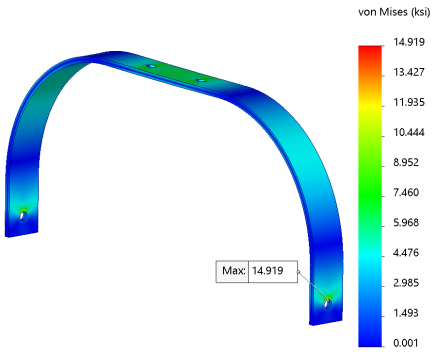


Figure 5.18: Rear Landing Gear Von Mises Stress

5.5.5 Banner Mechanism FEA

Following the identification of the drag force from banner towing, detailed in Section 4.4.7, a maximum critical drag load of 3 lbf was established. To validate the structural integrity of the rotary latch mechanism under this load, a static stress analysis was conducted using SolidWorks Simulation. The simulation focused on the worst-case loading Scenario, corresponding to the instantaneous "snatch" force during banner deployment (Stage II). The drag load was applied as a distributed bearing force on the contact surfaces of the rotary latch plates, simulating the interface with the tow line. The stationary plate was fully fixed at the servo mounting points to represent the rigid attachment to the fuselage.

The results indicated a maximum stress of 232 psi, located at the stress concentration of the latch hook. This value is significantly below the yield strength of the PLA material, resulting in a minimum factor of safety of 31.00. Furthermore, the deformation analysis confirmed that the total displacement of the rotary mechanism under load was negligible, ensuring that the mechanical interference required to retain the banner remains engaged until the servo is actuated. These results validate that the mechanism will maintain structural integrity and operational functionality throughout all three mission phases: stowing, high-load deployment, and release

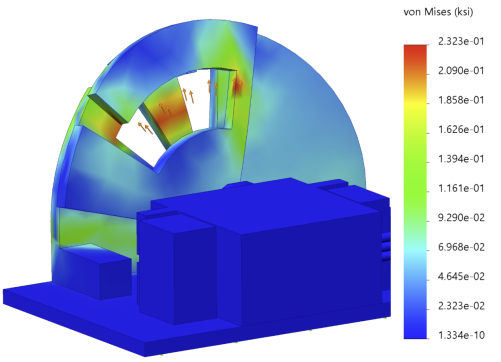


Figure 5.19: Banner Mechanism Von Mises Stress

5.6 Avionics

The aircraft avionics system provides regulated power distribution, radio control, pilot telemetry, and prototype flight data recording, with a summary of avionics and component weights provided in Table 5.3.

Avionics power is supplied by a 3-cell (3S) LiPo battery connected to a Spektrum RC Battery Eliminator Circuit (BEC) regulated to a nominal 6 V output. Although the avionics system requires only 6 V, the 3S battery provides increased voltage headroom, improving regulator efficiency and maintaining stable output during transient servo loading. Regulated power is distributed through a goBilda Servo Power Node (SPN), which serves as the primary power and signal distribution hub. The SPN splits the 6 V supply into six PWM signal channels with a shared power and ground bus, reducing wiring complexity and improving reliability.

The SPN interfaces with a FrSky X8R ACCST receiver, which distributes pilot command signals to the aircraft servos and receives control inputs from a FrSky Taranis QX7 transmitter. The X8R receiver was selected for its proven reliability and native telemetry support. Telemetry communication enables real-time transmission of system data from the aircraft to the transmitter, improving pilot situational awareness during flight.

The receiver Smart Port interface is connected to a FrSky Smart Port LiPo voltage sensor and FrSky TEMS

telemetry temperature sensor that monitor the propulsion battery. Propulsion pack voltage and temperature data are telemetrized to the transmitter display, allowing the pilot to monitor electrical and thermal conditions throughout flight without adding onboard control complexity.

Table 5.3: Aircraft Avionics Specifications

System	Component	Model	Mass (oz)
Telemetry and Control	Receiver 1	FrSky X8R	0.53
	Avionics Battery	Tattu 3S 11.1V 850mAh Lipo	2.50
	5 x Servos	KST DS113MG	0.77
	BEC	Hobbywing UBEC 10A	1.25
	Servo Controller	goBilda Servo Power Node	0.75
	Voltage Sensor	FrSky FLVS	0.26
	Temperature Sensor	FrSky TEMS	0.11
Banner Mechanism	Microcontroller	Arduino UNO R3	0.99
	Receiver 2	Spektrum SR315	0.21
	Servo	Spektrum A332	0.32
Flight Recording	Flight Controller	HolyBro Pixhawk 6C	1.3
	Airspeed Sensor	HolyBro MS4525DO	0.11
	GPS	HolyBro M10 GPS	1.10
Total			**10.2**

For prototype test flights, a dedicated flight recording system is installed that is fully independent of the primary avionics and not used during competition flights in accordance with competition rules. The flight recorder is centered around a Pixhawk 6C flight controller, which logs data using its embedded sensor suite, including a three-axis gyroscope, three-axis accelerometer, magnetometer, and barometric pressure sensor. A removable HolyBro M9N GPS module and HolyBro MS4525DO airspeed sensor are also connected during testing to record position, ground speed, and airspeed data. These sensors operate passively and do not provide control inputs to the aircraft.

5.7 Wiring

Figure 5.20 illustrates the wiring setup for onboard competition avionics and propulsion systems. WUDBF focused on streamlining wiring to reduce complexity, making the system easier to integrate, troubleshoot, and maintain.

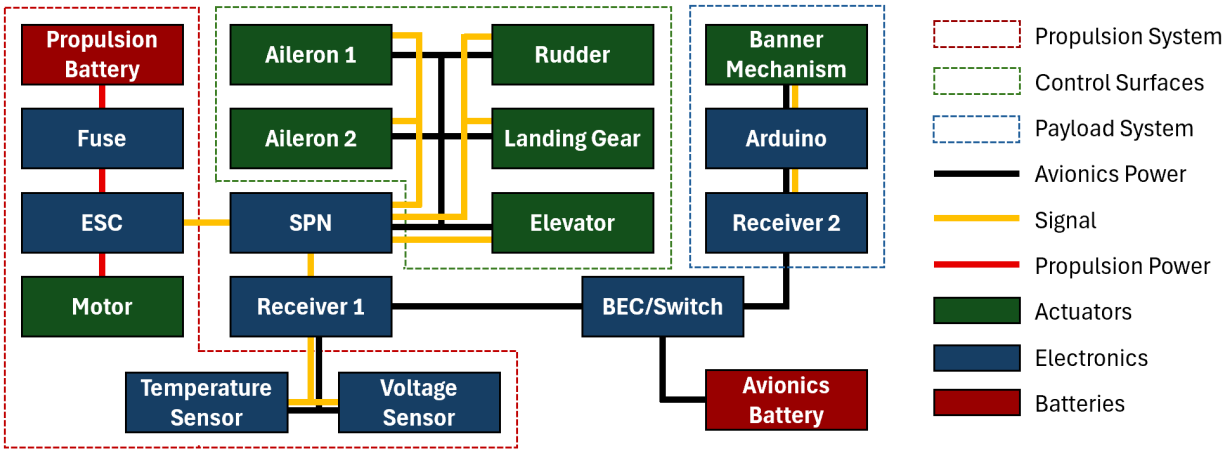


Figure 5.20: Wiring Diagram

5.8 Predicted Final Performance Score

Using the predicted aircraft performance summarized in Table 4.12, the final score was calculated using the mission performance equations detailed in Section 3.4. To obtain normalized scores, the maximum M2 score given in Equation 3.9 and the maximum M3 score given in Equation 3.11 were estimated based on the highest expected values achievable across all competing teams. M1 was not explicitly tabulated, as it is a completion-based mission and the aircraft is expected to successfully complete this mission. The resulting M2, M3, and GM scores, are summarized in Table 5.4.

Table 5.4: Predicted Mission Scoring Summary

(a) Mission 2

Mission 2			
Parameter	**Symbol**	**Value**	**Description**
Passengers Carried	$N_{passengers}$	16	Optimized value
Cargo Carried	N_{cargo}	4	Optimized value
Laps Completed	N_{laps}	8	Km_sim result
Battery Capacity	-	82 Wh	Optimized value
Total Score	M2	1.62	Eq. 3.9

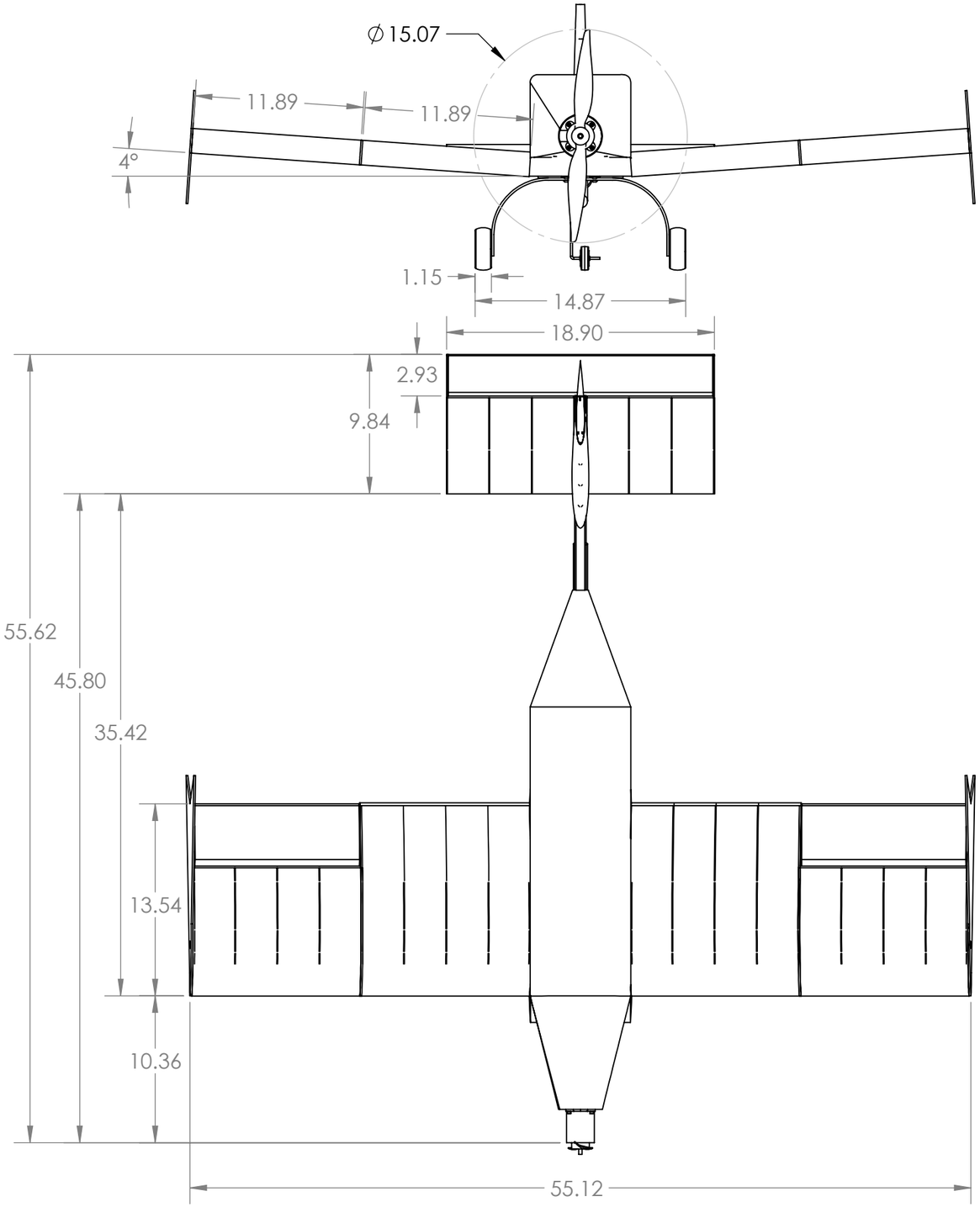
(b) Mission 3

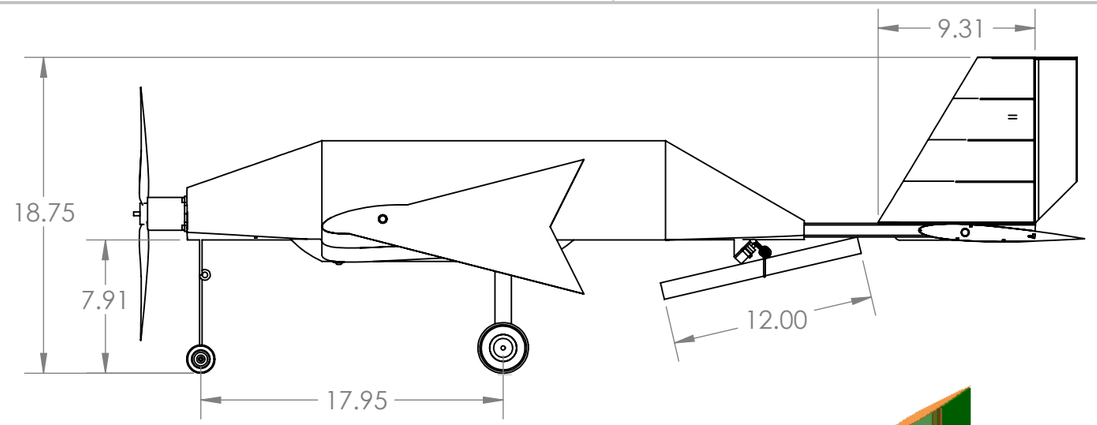
Mission 3			
Parameter	**Symbol**	**Value**	**Description**
Banner Length	-	60 in	Optimized value
Laps Completed	N_{laps}	6	Km_sim result
Wing Span	WS	4.59 ft	Optimized value
Total Score	M3	2.46	Eq. 3.11

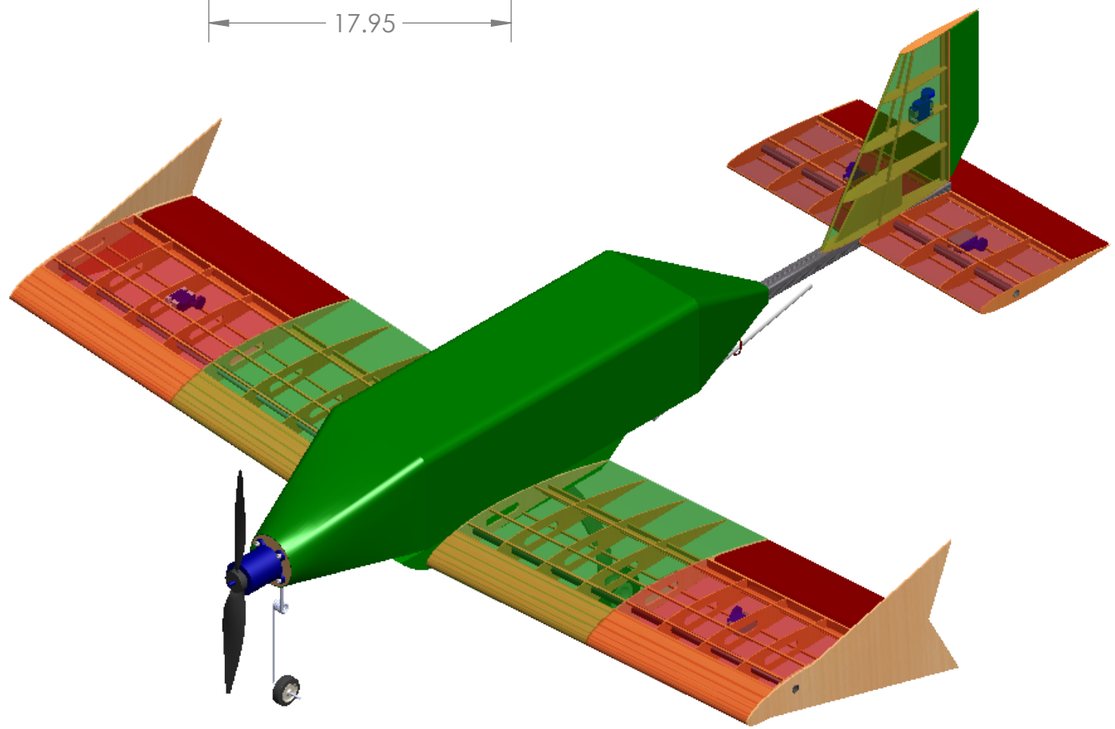
(c) GM

GM			
Parameter	**Symbol**	**Value**	**Description**
Completion Time	time	97 s	Tested value
Total Score	GM	0.39	Eq. 3.12

The predicted final score was calculated using Equation 3.3. Based on the individual mission scores summarized in Table 5.4, the aircraft is predicted to achieve a final score of **5.47** out of a maximum possible 8.0 points.







WUDBF	Drawing Title: 2025-2026 Design/Build/Fly Competition, WashU Psyduck	
Drawn by: Jacob Tenorio	Checked by: Sarah Donner	Date: 2/13/2026
All dimensions are in inches, unless otherwise specified.		
Size: B	Drawing: Three-view	Units: in
Scale: 1:8	Weight (lb): 7.68 empty	Sheet: 1 of 4

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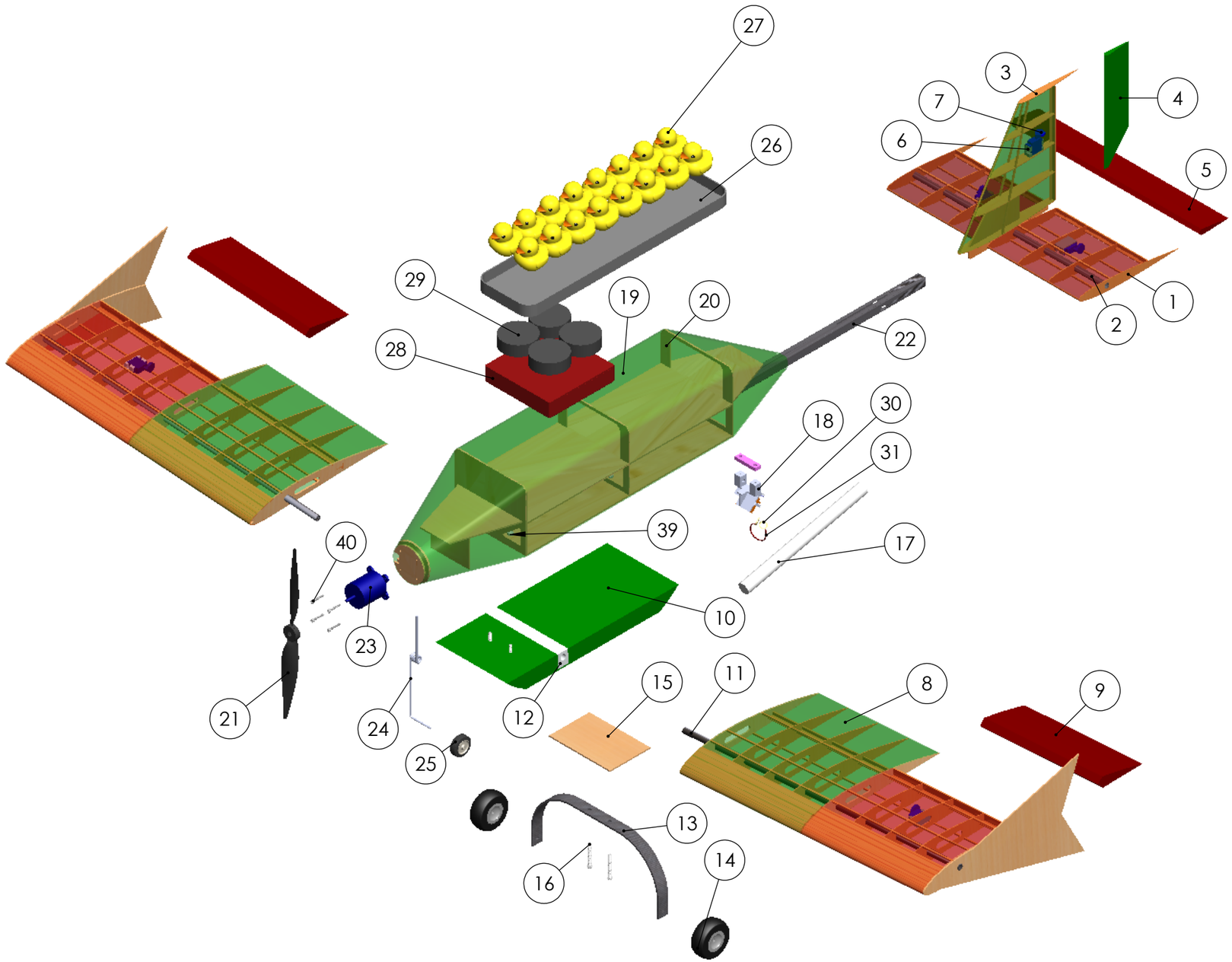
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Horizontal Stabilizer Ribs	Balsa / Plywood	1
2	Tail Spar	Carbon Fiber	1
3	Tail Vertical Ribs	Balsa / Plywood	1
4	Rudder	XPS Foam	1
5	Elevator	XPS Foam	1
6	Servo Mount	PLA	4
7	Servo	Plastic	4
8	Wing Ribs	Balsa / Plywood	1
9	Aileron	XPS Foam	2
10	Wingbox	XPS Foam	1
11	Wing Spar	Carbon Fiber	1
12	Wing Spar Bracket	Aluminum	1
13	Landing Gear	Carbon Fiber	1
14	Landing Gear Wheel	Rubber	2
15	Stability Plate	Plywood	1
16	Screws	Steel	4
17	Banner	13 oz. Vinyl	1
18	Banner Mechanism	PLA	1
19	Fuselage Shell	Fiberglass	1
20	Firewalls	Plywood	1
21	Propeller	Plastic	1
22	Tail Boom	Carbon Fiber	1
23	Motor	Aluminum	1
24	Front Landing Gear Torsion Coil	Steel	1
25	Front Wheel	Foam	1
26	Passenger Base Plate	PLA	1
27	Duck	Rubber	16
28	Cargo Restraint Box	PLA	1
29	Puck	Rubber	4
30	Washer	Steel	2
31	Rubber band	Rubber	1



DWG: Exploded View	Size: B	Scale: 1:8	Sheet: 2 of 4

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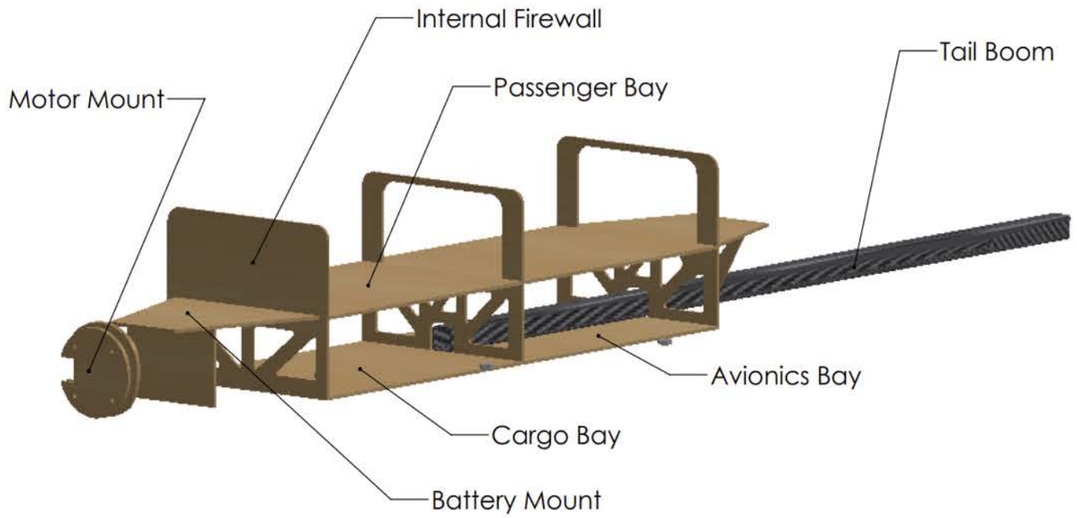
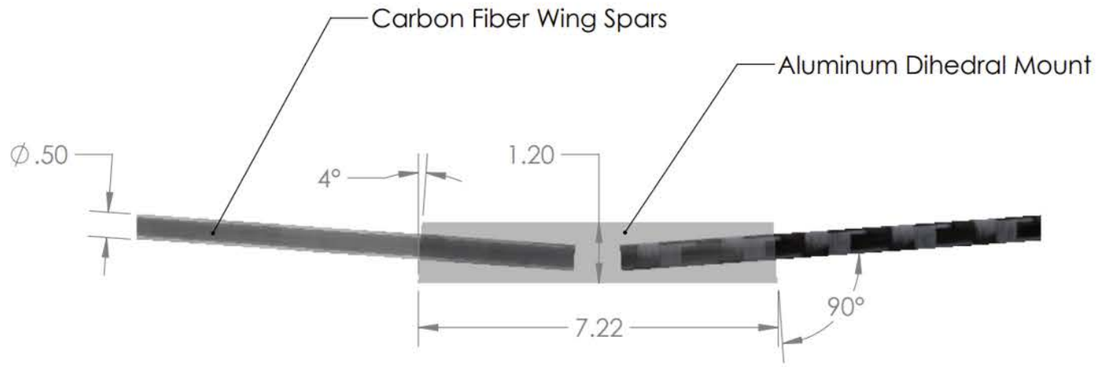
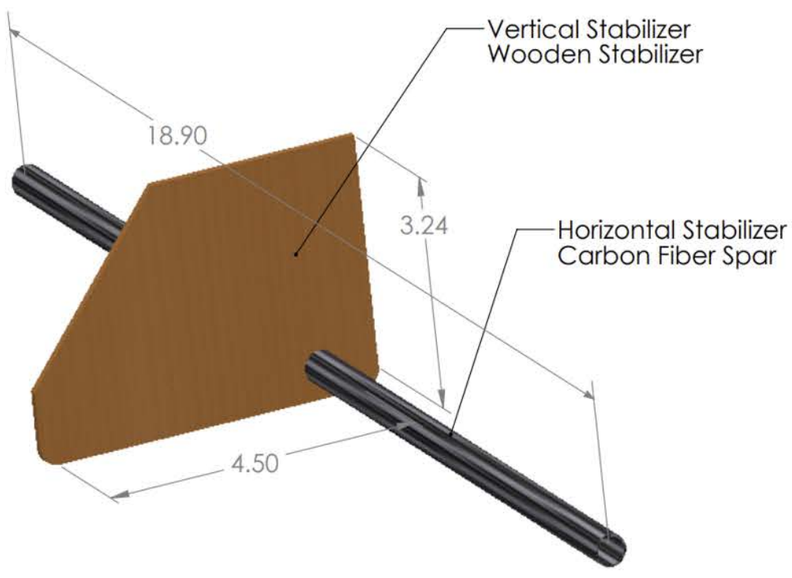
Internal Fuselage Geometry



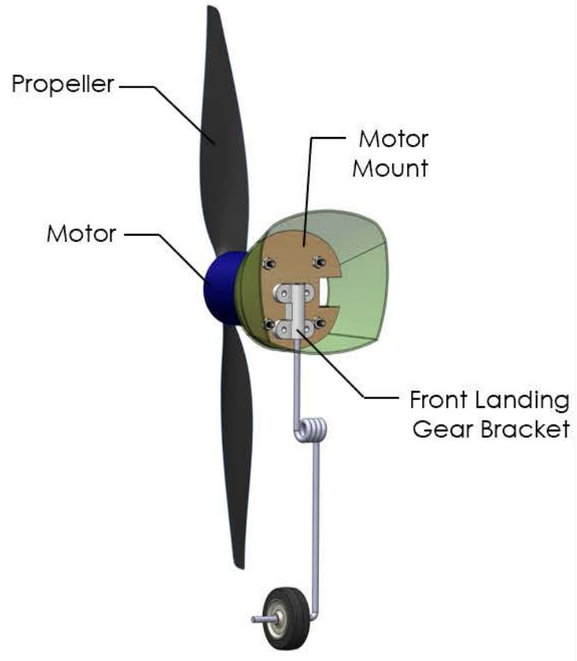
Dihedral Mount



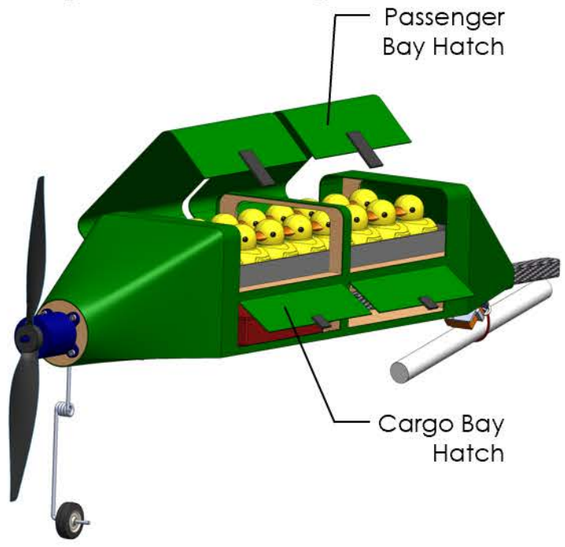
Empennage Stability Interface



Front Landing Gear



Fuselage Hatch Design



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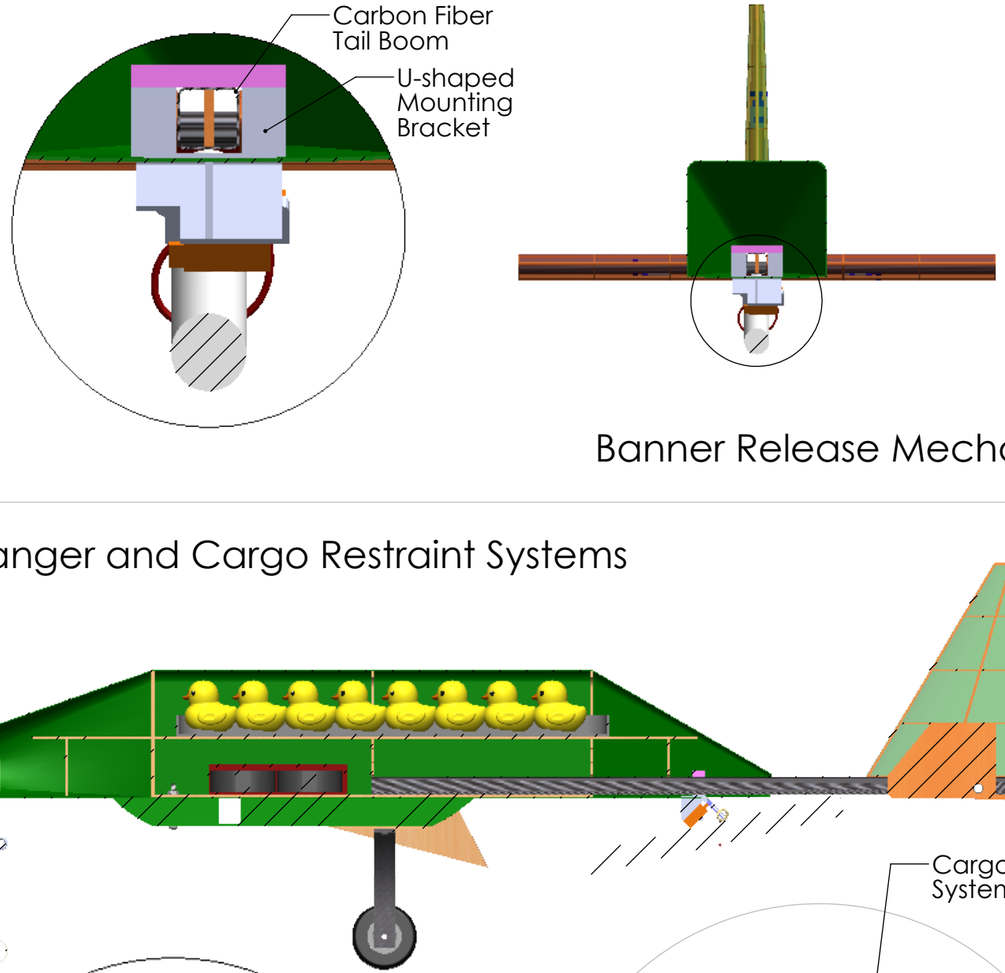
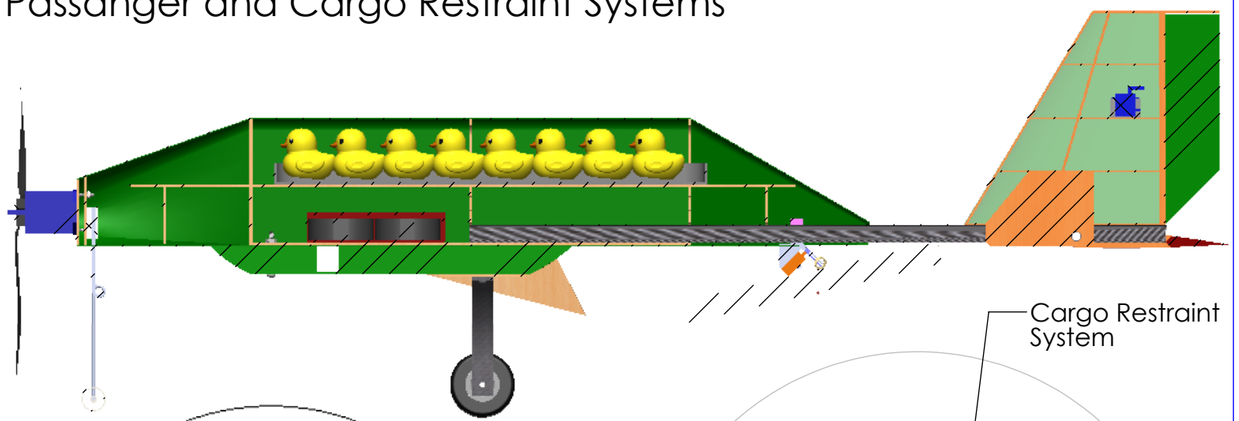
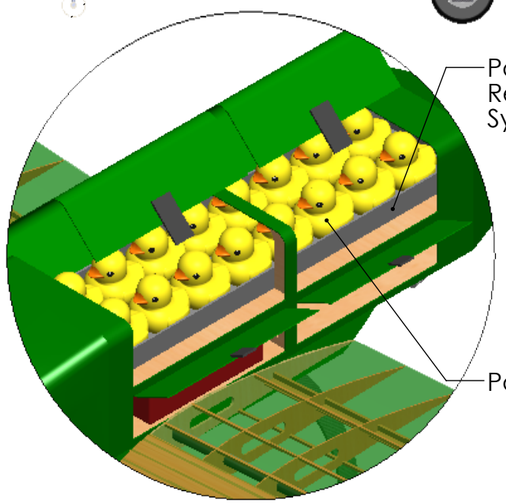
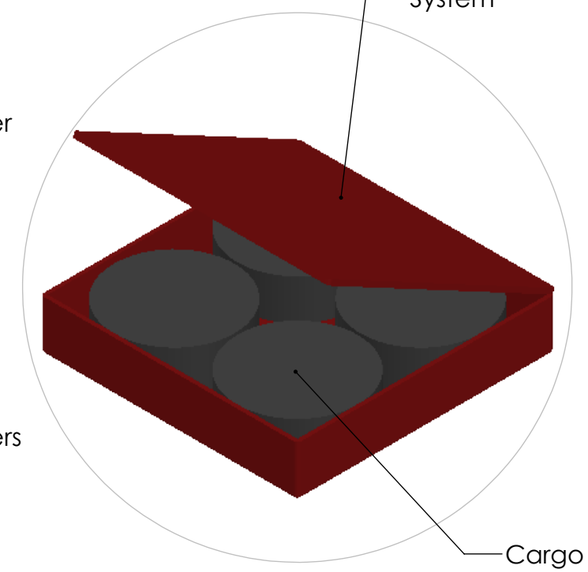


Banner Release Mechanism

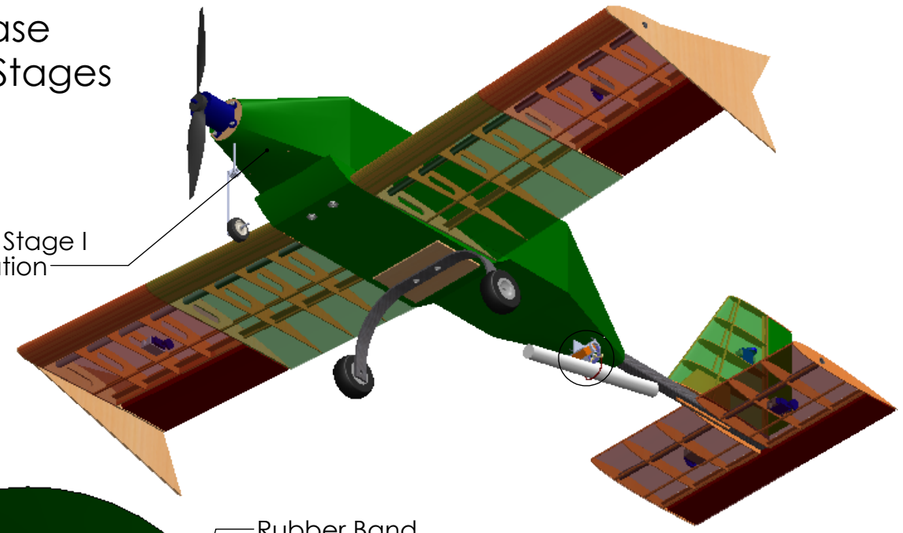
Passanger and Cargo Restraint Systems

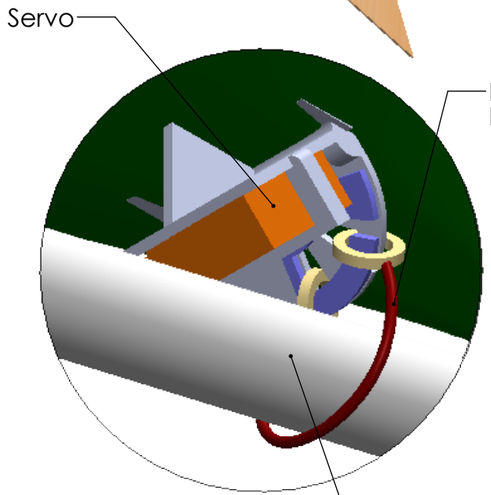




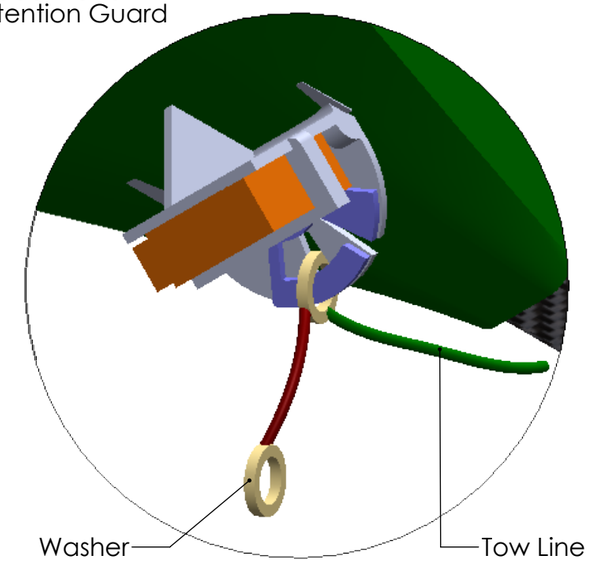


Banner Release Mechanism Stages

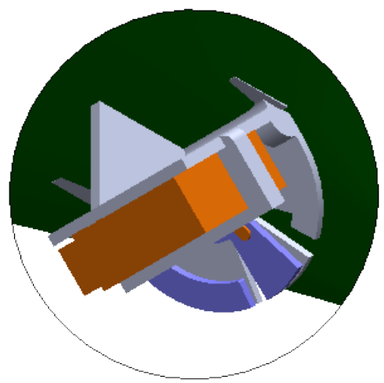




Stage I



Stage II



Stage III

DWG: Mission Configurations	Size: B	Scale: 1:8	Sheet: 4 of 4

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6 Manufacturing Plan

6.1 Manufacturing Process Evaluation

The materials and manufacturing techniques considered for the 2026 aircraft were informed by WUDBF's prior manufacturing experience, available resources, and the ability of each process to achieve the required manufacturing quality and aircraft performance.

6.1.1 XPS Foam Wire Cutting

XPS Foam core composite manufacturing using hot-wire cutting is useful for creating complex curvature with minimal manufacturing complexity and low material and tooling cost. XPS foam is lightweight, easy to shape, and well-suited for rapid iteration, making it suitable for parts requiring moderate strength and unique geometry. Wire-cut foam can be used for structural components as well as foam plugs used as molds for composite layups.

6.1.2 Additive Manufacturing

Additive manufacturing, primarily 3D printing, is the technique most suited for rapid iteration and producing complex geometries. Digitally designed parts can be quickly prototyped and refined, allowing design improvements to be implemented quickly throughout development. Lightweight Polyethylene Terephthalate Glycol (PETG) and PLA filament 3D prints are best for components with intricate features and tight integration requirements, but not for components with high strength requirements.

6.1.3 Composite Manufacturing

Composites are highly effective in engineering due to their high stiffness-to-weight ratio. Fiberglass reinforced epoxy laminates can be fabricated using a wet layup process over foam plugs and cured under vacuum. Composites can be used to coat components for reinforcement, or make structural components using molds. Composite manufacturing, however, requires increased fabrication time and careful quality control.

6.1.4 Wooden Construction

Wood construction has benefits including its low weight, ease of fabrication, and high repeatability, making it well suited for precision manufacturing. In remote controlled aircraft manufacturing, laser cutting is used to produce highly precise 2D parts to be assembled into 3D structures. Wood, depending on material properties, can be used for structural components, lightweight non-structural components, and skinning applications. Skinning involves dampening thin wood sections and forming them over molds to achieve curved geometries. Limitations arise from the anisotropic nature of wood, sensitivity to moisture, and warping during high-temperature laser cutting.

6.1.5 CNC Machining

CNC machining is useful for the fabrication of critical custom high load-bearing components due to its compatibility with high-strength metals, complex geometries, and high dimensional tolerances. CNC machining may require specialized equipment beyond in-house capabilities.

6.1.6 Monokote Wrapping

Monokote wrapping creates continuous aerodynamic surfaces over the structural wooden ribs, while adding insignificant weight. The film is a thin, flexible plastic that shrinks when heated, allowing it to reduce surface roughness and limit flow separation. While effective for lightweight surface formation, Monokote is susceptible to punctures, and its heat-induced shrinkage can locally deform thin or lightly supported structures if not carefully applied.

6.2 Selected Manufacturing Process

Each manufacturing process was chosen to best suit the structural, geometric, and load requirements of individual components. Cost, weight, complexity, and strength were considered when assessing and selecting materials.

6.2.1 Wings

Figure 6.1: Wing

The wing was constructed using a combination of wood construction, CF spars, composite control surfaces, and Monokote covering. Each half wing (left and right) were manufactured separately to allow for integration with the dihedral piece and foam belly fairing. Structural components, including ribs, jigs, plates, and spar supports, were laser cut to ensure consistent geometry and repeatability. Wooden components were designed to interlock, and were permanently bonded using epoxy and cyanoacrylate adhesive. Wing ribs were fabricated primarily from 1/16 in balsa, with 1/8 in birch plywood used for the outer ribs and ribs subjected to higher structural loads. Assembly jigs were used to position and align the ribs during construction, after which balsa backplates and stringers were installed to provide structural continuity and stiffness. A 1/4 in square balsa stringer was placed along the leading edge, with additional 1/8 in square balsa stringers installed at key rib locations. For the forward portion of the wing, 1/16 in balsa skins were dampened, formed over the rib and stringer framework, and bonded in place to define the leading-edge geometry. A CF spar was integrated through the rib assembly to serve as the primary load-bearing member for bending loads in flight. Servo mounting boxes were incorporated into the wing structure to provide localized reinforcement, shown in Figure 6.1. The completed wing structure was covered with heated Monokote film to form a lightweight, continuous aerodynamic surface. Wing control surfaces were fabricated separately using XPS foam cores shaped with a jigsaw and hot-wire cutter and reinforced with a single-ply fiberglass layup. Once each half-wing was full completed, the the respective CF rods on each side were set into the dihedral mount, shown in Figure 6.2, and set with epoxy.

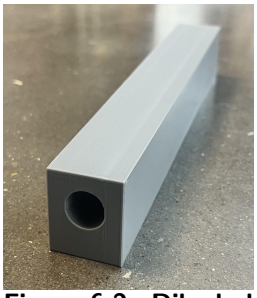
Figure 6.2: Dihedral Mount

6.2.2 Empennage

The horizontal and vertical tail surfaces were manufactured using the same materials and wood construction techniques as the wing. Laser-cut ribs and structural elements were fabricated primarily from 1/16 in balsa, with 1/8 in birch plywood used in regions subjected to higher loads. The horizontal stabilizer incorporated a CF rod to provide bending stiffness, while the vertical stabilizer utilized a birch wood wall as its primary structural member. The tail structures incorporated 1/8 in square balsa stringers as required. The rudder was constructed using XPS foam cores reinforced with a single-ply fiberglass layup. The elevator was fabricated from two balsa wood pieces bonded with epoxy and subsequently sanded to a tapered triangular profile. The control surfaces were mounted to the servos using the same attachment method as the wing control surfaces. The completed tail structures were covered with Monokote film and mounted into precision-cut slots in a CF boom passing through the fuselage, as shown in Figure 6.3.

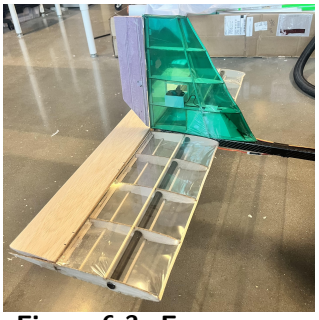
Figure 6.3: Empennage

6.2.3 Fuselage

The fuselage was fabricated using a combination of wooden construction, XPS foam hot-wire cutting, composite layup, and CNC machining. Primary internal fuselage structure, including the floor, firewalls, and motor mount, was manufactured from 1/8 in birch plywood and laser cut to the required geometry. Female molds were designed and hot-wire cut from XPS foam, into which the wooden structural

Figure 6.4: Fuselage Shell

components were installed to form a full-scale foam plug. A fiberglass-reinforced epoxy wet layup was then applied over the plug using a three-ply laminate in a $90°$–$45°$–$90°$ fiber orientation. The outer $90°$ plies utilized lighter-weight fiberglass fabric, while the central $45°$ ply used a heavier fabric. The fiberglass laminate was saturated with epoxy and cured under vacuum bagging. After curing, the internal foam plug was dissolved with acetone, resulting in a hollow composite fuselage shell as shown in Figure 6.4.

The fuselage was completed with the addition of a fiberglass-reinforced, hot-wire-cut XPS foam belly fairing, which housed a CNC-machined aluminum dihedral block. The dihedral block was bonded into the fairing cutout using epoxy.

Figure 6.5: Rear Landing Gear

6.2.4 Landing Gear

The landing gear assembly consisted of a commercially available CF rear landing gear, and aluminum front landing gear, both sourced from an established supplier. CF landing gear components are designed to withstand high impact and cyclic loads encountered during takeoff, landing, and ground operations, making them well suited for this application compared to in-house fabricated alternatives. The selected landing gear was integrated into the aircraft through attachment to the fiberglass reinforced foam belly fairing. To mitigate local stress concentrations at the mounting interface, a wooden backing plate and washers were incorporated to distribute landing loads over a larger area of the wing center section.

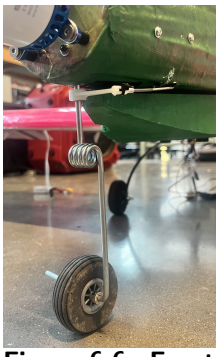

Figure 6.6: Front Landing Gear

6.2.5 M2 Restraint Systems

The passenger compartment was manufactured using fused deposition modeling with PLA filament. A diamond-cut polyester mesh was purchased and cut to size to restrain the ducks under slight tension. Snap fasteners were bonded at their respective locations to ensure proper alignment between the male and female components during installation of the mesh. Double-sided tape was then applied to fully line the base of the container to further limit payload motion. Finally, the container base was epoxied to the fuselage floor to prevent relative movement during ground handling and flight.

The cargo container and lid were similarly manufactured using PLA to maintain dimensional accuracy and minimize unneeded weight. A reinforced duct tape hinge was used to minimize part count and overall weight. The cargo is placed into the box outside of the aircraft, with the lid being closed with an additional strip of duck tape. The box is then installed into the fuselage and secured using hook-and-loop fasteners bonded to both the container and the fuselage structure to prevent movement during ground handling and flight.

6.2.6 M3 Payload

The banner was commercially sourced and hand-painted with the WashU logo. To ensure aerodynamic stability, a CF dowel was integrated into the leading edge for rigidity using hot glue, while aluminum washers were secured to the bottom corners to serve as ballast using fishing line. The towing interface consists of fishing line tied to the leading edge grommets using high-strength Uni knots.

The banner release mechanism features two rotary plates, additively manufactured from PLA, which were directly mounted to the servo arm. To secure the actuator, the servo itself was bonded to a 3D-printed PLA mounting plate using epoxy. The electronic interface was established by soldering the servo pins to a breadboard, which was then mechanically fastened to the CF tail boom and the fuselage shell to ensure structural continuity.

6.3 Manufacturing Milestones

A manufacturing schedule was implemented to ensure that components were constructed on time. The construction of the prototype and final aircraft were divided into important subtasks which were then used to create the schedule shown in Fig. 6.7 below.

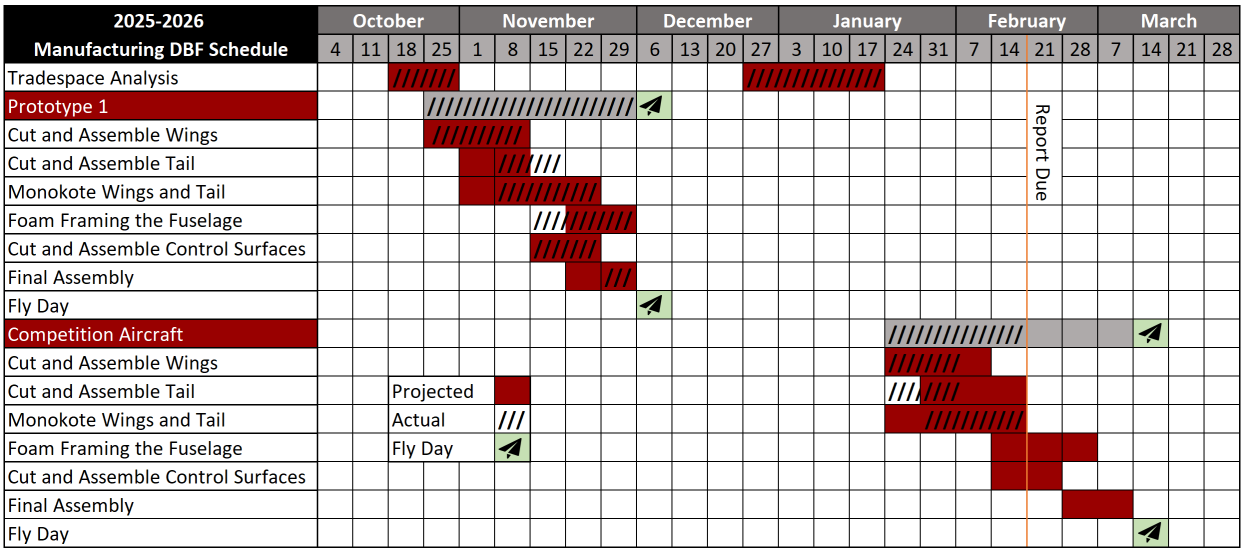

Figure 6.7: Manufacturing Schedule for Prototype and Competition Aircraft

7 Testing Plan

Testing verified that the aircraft met all subsystem-level requirements and was capable of safely and reliably completing the DBF competition missions. The test program progressed from component-level evaluations to subsystem tests and finally to full-aircraft ground and flight testing. minimizing integration risk. Each requirement in Table 3.7, the Subsystem Design Requirements Matrix, was validated.

Visual inspections and computational simulations are not listed as standalone tests in this section. Visual checks for rule compliance, structural fastening, and subsystem accessibility were continuously performed during manufacturing and throughout testing. Similarly, aerodynamic analysis, propulsion modeling, structural prediction, and mission-performance simulations were completed during the preliminary design. Section 7 focuses solely on the ground and flight tests under realistic operating conditions tabulated in Section 7.1.

7.1 Test Objectives and Schedule

The testing program was organized to validate all general (GR), operational (O), and payload (P) requirements through a structured sequence of ground, wind-tunnel, and flight tests. The overall objectives guiding this program are summarized in Table 7.1, which outlines the purpose of each testing category and its relationship to subsystem responsibilities.

Table 7.1: Testing Objectives

Sub Team	Test Goal	Requirements Verified
Ground Testing		
Aerodynamics + Payload	Ensure Mission 2 payload systems support rapid loading, secure restraint, and reliable removal under competition procedures.	O.1-3, O.9, P.1-4, P.7
	Validate mechanical reliability of banner stowage, deployment, towing connection, and release mechanisms prior to flight.	O.8, P.5-10
Manufacturing	Confirm landing-gear durability and load absorption capability during takeoff and landing operations.	GR.4, O.3
	Verify that all structural components, joints, and fasteners withstand expected mission loads and satisfy rule-required mechanical fastening criteria.	GR.4, GR.12–16, O.3
Propulsion	Validate propulsion system thrust output, power draw, and compliance with battery and ESC limits prior to flight.	GR.8–11, O.4–7
Wind Tunnel Testing		
Aerodynamics + Payload	Evaluate banner deployment reliability under controlled airflow conditions and identify aerodynamic sensitivities before flight testing.	O.8, P.5-8
	Confirm banner visibility, vertical orientation, stability, and rule-compliant readability under mission-level airflow.	P.9, P.10
Flight Testing		
Aerodynamics + Payload	Demonstrate safe airworthiness, controllability, stall margins, and handling qualities across the mission flight envelope.	O.4-7, GR.1–3
	Validate lift capability, controllability, and performance across Mission 2 payload configurations, including maximum payload loading.	O.1-3, O.6, O.9, P.1–4, P.7
	Confirm reliable banner deployment, towing stability, and release during Mission 3 conditions.	O.8, P.5–10
Aerodynamics\ Propulsion	Demonstrate mission-speed performance, turn capability, and energy usage for DBF lap timing.	O.4, O.5, O.7
Full Team	Validate complete mission execution by performing integrated M1→M2→M3 operations under realistic competition procedures.	All mission-related requirements

To ensure full requirement traceability, each planned physical test was mapped directly to the specific requirements it verifies. This is presented in Table 7.2, showing how the combined test set provides complete verification coverage.

Table 7.2: Test-Requirement Matrix

Test	Requirement Verified
Ground Testing	
Wing Structural Testing	GR.4 (wing tip load test), O.3 (landing intact), O.6 (lift capability at MTOW)
Fuselage Structural Testing	GR.4 (structural integrity), GR.5 (separate compartments), GR.6 (payload capacity), GR.7 (rigid doors), GR.12-16 (fastening, accessibility, RC switch, safety), O.3 (landing intact)
Static Thrust Testing	GR.8-11 (COTS propulsion, energy limit, fuse, safety), O.4 (M1 lap speed), O.5 (M2 lap maximization), O.7 (M3 lap capability)
Landing Gear Drop Testing	GR.4 (load capability), O.3 (structural landing), O.6 (supporting max payload during landing)
GM Testing	**GM / Operations:** O.1 (clean entry), O.2 (≤5 min assembly), O.3 (intact landing inspection), O.9 (GM speed) **M2 Payload:** P.1 (single-plane passengers), P.2 (individual boarding + vertical restraint), P.3 (separate cargo bay), P.4 (cargo restraint) **Banner Handling:** P.7 (external stowage, no interference), P.8 (remote deployment/release demonstrated in GM), P.10 (basic durability during handling)
Wind Tunnel Testing	
Banner Deployment Testing	O.8 (deployment under airflow), P.5 (length/aspect ratio), P.6 (logo visibility prep), P.7 (stowage clearance), P.8 (deployed correctly under airflow), P.9 (vertical orientation initiation)
Banner Visibility and Stability Testing	O.8 (stable towing behavior pre-flight), P.6 (logo/readability), P.9 (vertical orientation), P.10 (no damage)

Flight Testing	
M1 Simulation	GR.1–3 (configuration), O.3 (structurally intact landing), O.4 (3 laps in ≤5 minutes), handling qualities required for all missions
M2 Simulation	O.5 (maximize laps), O.6 (fly with maximum payload), P.1–4 (passenger/cargo integration), P.7 (banner stowage during M2), O.3 (landing intact), O.9 (GM–flight consistency)
M3 Simulation	O.7 (lap performance), O.8 (banner deploy + release in air), P.5–10 (banner geometry, visibility, durability, orientation), O.3 (landing intact)
Full Competition Simulation (M1 → M2 → M3)	ALL mission-level requirements: O.1–9(all payload requirements), P.1–10(configuration), GR.1–G3(structural), GR.4(operational integrity), GR.5–16 (as exercised across missions)

The schedule of these tests are detailed in Figure 7.1.

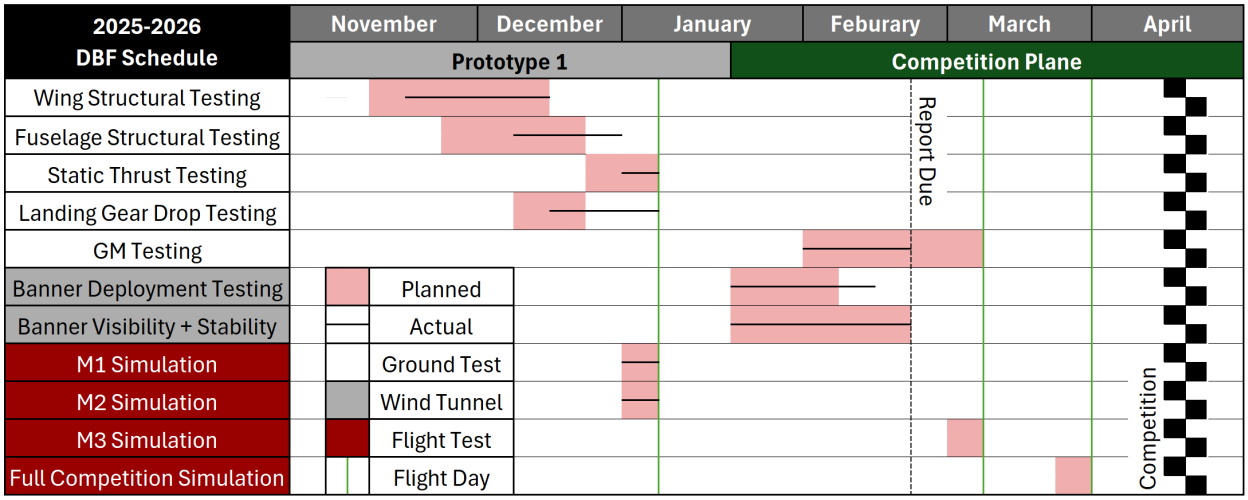

Figure 7.1: Testing Schedule

7.2 Ground Testing

7.2.1 Wing Structural Testing

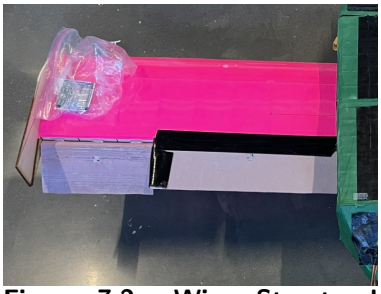

Figure 7.2: Wing Structural Testing

Wing structural testing was conducted to satisfy the structural load verification objective by demonstrating that the wing could safely sustain maximum flight loads at the aircraft's maximum takeoff weight prior to flight testing. The aircraft was configured at MTOW is 10.88 lb. To replicate the critical wing bending condition, a downward load of 7 lb was applied at each wing tip. Wing deflection was measured at multiple spanwise locations, with a maximum tip deflection of 0.15 in recorded during testing. The wing spars, attachment points, and control surfaces were monitored throughout the test for structural damage.

The wing structure remained intact and returned to its original unloaded geometry following load removal. Successful completion of this test verified compliance with wing load capability and lift-related operational requirements, satisfying the structural verification objective defined in the testing program.

7.2.2 Fuselage Structural Testing

Fuselage structural testing was performed to confirm that the fuselage and integrated payload compartments could withstand expected ground, landing, and in-flight loads without structural damage or loss of compartment separation. The fully assembled aircraft was shaken by hand and rotated through multiple orientations for M2

and M3 flight configurations.

During testing, the passenger and cargo payloads remained fully restrained, and no relative motion of the ducks or hockey pucks was observed. The banner deployment mechanism also remained securely attached and fully functional throughout testing. Post-test inspection verified that the fuselage structure, bulkheads, access doors, and fasteners remained intact, with no cracking or permanent deformation observed, demonstrating compliance with structural integrity, payload retention, and intact landing requirements.

7.2.3 Static Thrust Testing

Static thrust testing was performed on the chosen propulsion system using an RCbenchmark Series 1585 Dynamometer mounted to a thrust stand. Using RCbenchmark software, thrust, RPM, current draw, and electrical power measurements were collected at incremental throttle levels. The data was then used to validate predicted performance and confirm the propulsion system met the necessary requirements.

7.2.4 Landing Gear Drop Testing

The landing gear was identified as a critical structural risk due to the aircraft's high MTOW of 10.88 lb and elevated landing speed in M2. The landing speed was assumed to be approximately equal to the takeoff speed, the aircraft would experience an abrupt landing, imparting significant compressive loads on the main and nose landing gear.

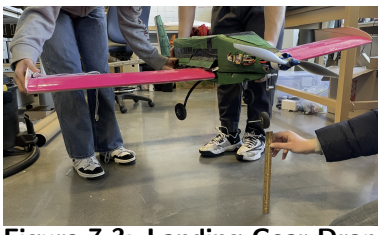

Figure 7.3: Landing Gear Drop Testing

To verify that the landing gear could sustain these loads, the aircraft was subjected to controlled drop testing at the MTOW configuration. The aircraft was dropped from a height of 12 in, corresponding to an equivalent descent rate of 50 ft/s, representative of a hard landing condition. The drop tests were recorded and used to measure landing gear deflection, as shown in Figure 7.3.

No cracking, plastic deformation, or fastener loosening was observed, and the landing gear remained fully functional after testing. These results verified that the landing gear is capable of sustaining landing loads at maximum takeoff weight and satisfies intact landing and load capability requirements.

7.2.5 GM Testing

Ground Mission timing tests were conducted to evaluate the loading, installation, and operation of all ground mission components and to identify potential bottlenecks for future optimization. Testing was performed within a 10 ft by 10 ft mission box using the fully assembled aircraft, and each stage of the ground mission was completed and timed multiple times to determine an average completion time of 97 seconds.

Each test iteration included installation and restraint of passengers and cargo, verification of control functionality, installation of the banner in the stowed configuration, and commanded deployment and release of the banner. Testing confirmed that payload installation could be completed efficiently, restraints remained secure, and the banner mechanism operated reliably under time pressure, demonstrating compliance with ground mission and payload handling requirements.

7.3 Wind Tunnel Testing

Wind tunnel testing was conducted to support the aerodynamic validation objectives identified in the testing objectives Table 7.1. Testing was performed using the low-speed wind tunnel at Saint Louis University to isolate banner aerodynamic effects independent of full-aircraft interactions.

7.3.1 Banner Deployment Testing

Banner deployment testing was conducted at the aircraft's nominal cruise speed of 105 ft/s to evaluate the functional reliability of the banner deployment and jettison mechanism under flight-representative conditions. The complete mechanism was installed to the roof of the wind tunnel, and repeated deployment and release cycles were performed to assess operational consistency.

7.3.2 Banner Visibility and Stability Testing

Banner stability and visibility testing was conducted in the wind tunnel over a freestream velocity range of 70 ft/s to 90 ft/s. Higher speeds were not tested due to structural limitations of the banner mounting hardware. Multiple banner configurations with lengths ranging from 30 in to 75 in were evaluated while maintaining a constant AR and tow-line geometry. Select configurations incorporated variations in mass distribution and surface slits to assess their influence on aerodynamic behavior. All banners were fabricated using the same materials and manufacturing methods described in Section 6.2.6 to ensure consistency across test articles.

The tested banners possessed significantly greater surface area than the analytically optimized configuration identified in Section 4.4.7. This deviation was intentional, as M3 requires a banner that remains visually detectable and dynamically stable at flight-relevant speeds. Prior research and preliminary flight observations indicated that banners near the analytically optimized area were insufficiently stable or visible during flight. Consequently, larger banner configurations were evaluated despite the associated increase in aerodynamic drag.

Testing focused on observing banner aerodynamic behavior under steady freestream conditions. Drag was quantified using an in-line force gauge with the banner constrained to limit large-scale motion, allowing validation of analytical drag estimates and assessment of relative trends with banner size and configuration.

Banner orientation stability and visibility were then evaluated under the same flow conditions by adjusting mass placement along the tow line and leading edge. Surface slits were also assessed for their effect on unsteady motion. Configurations were compared to identify the minimum added mass required to maintain consistent visibility while limiting aerodynamic and weight penalties.

7.4 Flight Testing

A test flight was conducted to verify that the aircraft met preliminary design goals and to provide an initial assessment of flight performance and handling qualities. The aircraft was instrumented with the onboard flight recording system described in Section 5.8, which was used to capture relevant flight data during testing. This data was analyzed to extract key performance metrics including airspeed, lap timing, and overall flight behavior. Pilot feedback was also incorporated to qualitatively assess aircraft stability, control response, and handling characteristics.

The completed flight testing included operation in both the M1 configuration (empty aircraft) and the M2 configuration with 16 passengers and 4 cargo payloads installed. At the time of report submission, a single flight test session had been completed. Additional ground, wind tunnel, and flight tests are planned to fully validate aircraft performance, payload capability, and mission execution, as detailed in Table 7.1. Results from the completed flight test and the planned testing campaign will be used to further refine the aircraft configuration and operating procedures leading into the competition.

7.5 Ground Inspection and Pre-Flight Checklists

WUDBF utilized the following checklists to ensure safety and correct documentation of flight data are met before all tests.

Table 7.3: Ground Inspection Checklist

Ground Inspection	
Assembly	
Wing attachment secure; no cracks or looseness	☐
Tail surfaces secure and aligned	☐
Landing gear secure; wheels rotate freely	☐
Control horns, clevises, and linkages secure with retainers	☐
No visible structural damage or delamination	☐
Avionics	
Correct control directions verified	☐
Failsafe verified	☐
Propulsion	
Propeller secure and undamaged	☐
ESC and wiring secure; no exposed conductors	☐
Battery installed and restrained	☐
Payload (if applicable)	
Correct number of passengers and cargo installed and restrained	☐
Banner stowed externally and secure	☐

Table 7.4: Flight Inspection Checklist

Flight Inspection	
Pre-Flight	
Payload (if applicable) secured	☐
Propulsion battery level check	☐
Avionics battery level check	☐
CG Position Confirmed	☐
Wingtip test	☐
Reciver turned on	☐
Control surfaces test	☐
Taxi test	☐
All doors and hatches securly closed	☐
Post-Flight	
Fuse in	☐
Propulsion battery level check	☐
Avionics battery level check	☐
Visual Inspection	☐

8 Performance Results

8.1 Ground Test Results

8.1.1 Static Thrust Test Results

Results from static thrust testing at maximum performance (100% throttle) confirmed that the propulsion system is able to meet the required performance targets. A comparison between measured values and eCalc predictions is presented in Table 8.1.

Based on the static testing, the motor drew slightly more burst current than predicted. This is not an issue, however, since the ESC was sized with significant margin. In addition, the static thrust appears greater than what was expected due to ecalc's predicted 10% +/- error. Later flight testing confirmed that the propulsion system delivered the desired takeoff capability and cruise performance.

Table 8.1: Predicted vs. actual performance.

Parameter	Predicted	Test Result
RPM	8690	9223
Thrust (lbf)	15.12	15.02
Max Current (A)	63.30	51.70
Electrical Power (W)	1589	1428

The measured propulsion thrust was slightly lower than predicted, likely due to ground effects and airflow obstruction inherent to static testing. Subsequent flight testing confirmed that the propulsion system provided adequate take-off and cruise performance for M1 and M2.

8.2 Wind Tunnel Test Results

8.2.1 Banner Deployment Test Results

Across five deployment attempts, the system successfully deployed and jettisoned the banner in four cases, corresponding to an 80% success rate. In the single unsuccessful attempt, the washer interface between the first and second stages of the mechanism became lodged, preventing full deployment. No partial releases or unintended

banner separations were observed.

These results indicate that the deployment concept is fundamentally sound but sensitive to component tolerances at the stage interfaces. Planned refinements include adjusting slot dimensions and washer sizes to improve alignment margins and reduce the likelihood of mechanical interference prior to competition flights. Subsequent testing focused on banner aerodynamic behavior, stability, and visibility once deployed.

8.2.2 Banner Visibility and Stability Test Results

Table 8.2: Banner Drag Force Results

Banner Dimensions (in)	Added Weight (lb)	Slits	Max Speed (ft/s)	Drag (lbf)
6 x 30	0.105	no	87	6.6
8 x 40	0.105	no	81	3.7
10 x 50	0.105	no	81	4.5
12 x 60	0.195	no	81	8.9
12 x 60	0.195	yes	81	6.4
15 x 75	0.195	no	73	8.0
15 x 75	0.195	yes	73	5.7

The measured banner drag did not follow the expected linear scaling with projected area predicted by the analytical model in Section 4.4.7. Smaller-area banners exhibited significantly higher drag than anticipated, which was attributed to pronounced dynamic instability during testing. These banners frequently oscillated, twisted, and intermittently collapsed, producing unsteady flow separation and impulse-like force spikes that increased the measured steady-state drag. As banner area increased, stability improved and force measurements became more repeatable. No tearing or material damage was observed for any banner configuration throughout testing.

For both the 12 in by 60 in and 15 in by 75 in banners, the introduction of surface slits resulted in a drag reduction of approximately 33% while also significantly improving banner visibility by reducing unsteady motion and promoting a consistent vertical orientation. The most visible configuration observed during testing was the 12 in by 60 in banner with four slits, and this configuration was therefore selected as the baseline for M3.

Although the drag associated with this configuration remains higher than originally predicted and was previously identified as potentially reducing the aircraft lift-to-drag ratio below acceptable levels, several mitigation strategies are planned. These include continued optimization of mass distribution along the tow line and leading edge to improve stability, refinement of slit number and placement to reduce pressure buildup, and adjustment of M3 flight speed to balance visibility and aerodynamic performance. These approaches will be evaluated through additional wind tunnel testing and full-scale flight testing to ensure Mission 3 completion without compromising overall aircraft performance.

8.3 Flight Testing

Dec. 6th, 2025 - Multiple Flights (Prototype)

Psyduck performed two flights during the Dec. 6th test flight of the first prototype. These flights consisted of a M1 simulation in the empty aircraft configuration and a M2 simulation with representative passenger and cargo payloads installed. The preliminary design configuration flown during this test is shown in Fig. 8.1.

Figure 8.1: Psyduck on Flight 2 of Dec. 6 Test Flights

Overall, the aircraft demonstrated stable and predictable handling characteristics throughout all test flights. Pitch and roll response remained crisp even in the fully loaded M2 configuration, and thrust response was determined to be more than adequate for all flight segments. These observations support the longitudinal and lateral stability predictions developed during the preliminary design phase and indicate sufficient control authority under maximum payload conditions.

For each test flight, WUDBF utilized an onboard airspeed sensor to record average and maximum airspeeds achieved during Mission 1 and M2 across key flight segments, including the 180° turn, 360° loop, and straight cruise. These values, summarized in Table 8.3, were critical for validating aerodynamic and propulsion predictions developed during the conceptual and preliminary design phases. Measuring airspeed over specific segments allowed for direct comparison to constraint sizing results and provided insight into possible aircraft improvements.

Table 8.3: Airspeed Data M1/M2

	M1		M2	
	Average Speed (ft/s)	Max Speed (ft/s)	Average Speed (ft/s)	Max Speed (ft/s)
180° Turn	84	105	83	102
360° Loop	76	99	75	96
Straight Cruise	87	130	81	123

Constraint sizing results presented in Table 4.12, the predicted straight cruise velocity was 115 ft/s for M1 and 114 ft/s for M2 at an estimated throttle setting of 70%. Flight testing revealed maximum straight cruise velocities of 130.13 ft/s for M1 and 122.97 ft/s for M2. These results represent performance increases of 13.2% and 7.9%, respectively, relative to analytical predictions.

According to data logging, these speeds were achieved at approximately 50% throttle, indicating that the propulsion system provided greater thrust margin than anticipated or that parasitic drag was underestimated during preliminary sizing.

The anticipated increased performance of the aircraft suggests a high competition score as a result of the under-

estimated overall aircraft speed.

The airspeeds over the banked turn and loop, along with yaw rate measured by the internal flight computer gyroscope, were used to calculate the turn radius of the aircraft. These values are detailed in Table 8.4. Calculating the turn radii inform conclusions on previously unknown load factors throughout turns.

Table 8.4: Calculated Turn Radius M1/M2

	M1	M2
Turn Radius for 180° Turn (ft)	167	182
Turn Radius for 360° Loop (ft)	154	175

Overall, the test flight data quantitatively validate the aerodynamic and propulsion models used in earlier design phases. The aircraft not only met predicted performance targets but demonstrated a higher capability, providing confidence in its ability to meet and exceed mission scoring predictions.

The 2025-26 WUDBF Team, shown in Figure 8.2, is excited to participate in the 2026 AIAA DBF Fly-Off with Psyduck!

Figure 8.2: WUDBF 2026 Team

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